The vacuum Bell correlation of the massive free scalar field attains the classical bound for bounded regions beyond a finite separation

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For the real free scalar field in 3 + 1 dimensions we prove that a universal finite $C > 2$ exists: for every mass $m > 0$ and radius $R > 0$, the vacuum Bell correlation of two equal-time double cones of radius R equals its classical bound at centre separations $L \geq CR$. The supremum defining the Bell correlation includes all self-adjoint local contractions; the vacuum remains entangled. By isotony, CHSH violation is absent for fixed bounded regions moved sufficiently far apart along a fixed spacelike direction. This resolves the fixed-bounded-region version of Open Quantum Problem 12.

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I. STATEMENT OF THE RESULT

Let O_1, O_2 be spacelike separated regions of Minkowski space, with local von Neumann algebras $\mathfrak{A} = \mathfrak{A}(O_1)$ and $\mathfrak{B} = \mathfrak{A}(O_2)$, which commute, and let ω be a state. An observable measurable in O_1 whose values lie in $[-1, 1]$ is a self-adjoint $A \in \mathfrak{A}$ with $\|A\| \leq 1$, a self-adjoint *contraction* ($\|\cdot\|$ is the operator norm). The CHSH inequality [1] concerns two such observables on each side. Following Summers and Werner [2], the *Bell correlation* of ω is

$$\beta(\omega; \mathfrak{A}, \mathfrak{B}) = \sup \tfrac{1}{2}\big|\omega\big(A_0(B_0 + B_1) + A_1(B_0 - B_1)\big)\big|, \quad (1)$$

the supremum over self-adjoint $A_i \in \mathfrak{A}$ and $B_j \in \mathfrak{B}$ of norm at most 1. The CHSH inequality is the statement that in a *local hidden variable model*, in which the four measured values are functions $a_i(\lambda), b_j(\lambda) \in [-1, 1]$ of a shared classical variable λ and expectations are averages over λ,

$$\tfrac{1}{2}|\langle a_0 b_0\rangle + \langle a_0 b_1\rangle + \langle a_1 b_0\rangle - \langle a_1 b_1\rangle| \leq 1, \quad (2)$$

because $|a_0(b_0 + b_1) + a_1(b_0 - b_1)| \leq |b_0 + b_1| + |b_0 - b_1| \leq 2$ for every λ. Quantum mechanics allows values of β up to $\sqrt{2}$ [3]: for *reflections*, observables with $A_i^2 = I$ and $B_j^2 = I$ and hence with outcomes ± 1, one has $\big(A_0(B_0 + B_1) + A_1(B_0 - B_1)\big)^2 = 4I - [A_0, A_1][B_0, B_1]$, of norm at most 8. The choice $A_i = B_j = I$ gives 1, so $\beta \geq 1$ always. If $\beta > 1$, the correlations of ω between the two regions cannot be reproduced by any local hidden variable model; this is a *Bell violation*. Note that the supremum in (1) is unchanged if it is restricted to reflections, because every self-adjoint contraction is an average of reflections and the expression is affine in each argument (Appendix A, Step 1). (Verch and Werner [4] omit the factor $\frac{1}{2}$, so that their bounds are 2 and $2\sqrt{2}$.)

By the Reeh–Schlieder theorem the vacuum is entangled across any two spacelike separated regions [5], and the entanglement is distillable at any distance [4]: local operations and classical communication can extract standard entangled pairs from it. Bell correlation is a stronger property. Summers and Werner proved $\beta = \sqrt{2}$ for the vacuum of a massive free field on complementary wedges ($\{x^1 > |x^0|\}$ and its spatial mirror) and on tangent double cones [6–9], and for regions at a positive distance a mass gap $m > 0$ gives $\beta \leq 1 + 2e^{-m\,t_{12}}$, where t_{12} is the largest time through which one region can be shifted while the two stay spacelike separated [2, 10]. This is an upper bound; whether some violation survives at every distance is Problem 12 of the IQOQI Vienna Open Quantum Problems list [11, 12]: "Decide whether some (necessarily small) violation of the inequalities is possible for regions arbitrarily far apart. For definiteness, consider a massive scalar free relativistic Bose field." Violations obtained after local filtering of coupled probes [13, 14] concern a post-selected state and are not at issue here. Nor is entanglement itself: suitable local modes detect vacuum entanglement at every distance [15], but entanglement of the unfiltered vacuum does not imply a CHSH violation.

Setting.—Units $\hbar = c = 1$, signature $(-, +, +, +)$, so that $x \cdot y = -x^0 y^0 + \mathbf{x} \cdot \mathbf{y}$ for points $(x^0, \mathbf{x})$ with $\mathbf{x} \in \mathbb{R}^3$; $B_R(\mathbf{x}_0) = \{\mathbf{x} \in \mathbb{R}^3 : |\mathbf{x} - \mathbf{x}_0| < R\}$ is the open ball of radius R; Fourier transforms are $\hat{f}(\mathbf{p}) = \int e^{-i\mathbf{p}\cdot\mathbf{x}} f(\mathbf{x})\, \mathrm{d}^3x$, and $\mathbf{e}_1 = (1, 0, 0)$. For $R > 0$ the double cone $D_R = \{(x^0, \mathbf{x}) : |x^0| + |\mathbf{x}| < R\}$ has base $B_R(0)$ at time zero, and $D_R + (0, L\mathbf{e}_1)$ has its centre at distance L along the first axis. We consider the real free scalar field of mass $m > 0$ in 3 + 1 dimensions, its vacuum state ω,

and its local algebras $\mathfrak{A}_m(O)$. For $L > 2R$ the two bases are disjoint, the algebras of the two double cones commute, and we write

$$\beta_m(R, L) = \beta\big(\omega; \mathfrak{A}_m(D_R), \mathfrak{A}_m(D_R + (0, L\mathbf{e}_1))\big).$$

The supremum in (1) runs over all bounded observables of the two regions; they may depend on L and may involve events of arbitrarily small probability.

Theorem 1 (Finite separation cutoff)**.** *There is a finite constant $C_* > 2$, independent of m and R, such that*

$$\forall\, m, R > 0, \qquad L \geq C_* R \quad \Longrightarrow \quad \beta_m(R, L) = 1. \tag{3}$$

Corollary 1 (Fixed bounded regions)**.** *For every $m > 0$, any two bounded regions O_1, O_2 and any fixed spacelike direction, the vacuum Bell correlation of $\mathfrak{A}_m(O_1)$ and of the local algebra of the translate of O_2 equals* 1 *for all sufficiently large translations of O_2 in that direction.*

This answers Problem 12 in the negative for fixed bounded regions. The theorem supplies a finite sufficient separation, not a sharp threshold: it does not locate the smallest such separation, and for regions that grow with the separation, and for wedges at a positive gap, the question remains open. The theorem is an existence statement, and the constant C_* it produces is finite but not explicit. What the theorem excludes is a CHSH violation, however small and however rare its outcomes, between fixed bounded regions far enough apart.

The proof compares the vacuum with the product of its restrictions to the two regions: the state with the same local physics and no correlations between the regions at all. Section III turns closeness to this product state into the classical bound $\beta = 1$, and Sections IV and V establish the closeness at large separation, uniformly in the mass.

Section II sets up the field and makes two reductions; Sections III–VI prove Theorem 1.

II. SETTING AND TWO REDUCTIONS

At time zero the field is described by the smeared field and momentum

$$\phi(f) = \int \phi(0, \mathbf{x}) f(\mathbf{x})\, \mathrm{d}^3x, \qquad \pi(g) = \int \dot\phi(0, \mathbf{x}) g(\mathbf{x})\, \mathrm{d}^3x,$$

$f, g \in C_c^\infty(\mathbb{R}^3; \mathbb{R})$, where $[\phi(0, \mathbf{x}), \dot\phi(0, \mathbf{y})] = \mathrm{i}\delta^3(\mathbf{x} - \mathbf{y})$. We call $\mathsf{f} = (f, g)$ a *source*, denoted by the sans-serif letter of its field component, and write

$$\Phi(\mathsf{f}) = \phi(f) + \pi(g) \tag{4}$$

and $\mathsf{W}(\mathsf{f}) = e^{\mathrm{i}\Phi(\mathsf{f})}$. The commutator is $[\Phi(\mathsf{f}), \Phi(\mathsf{f}')] = \mathrm{i}\varsigma(\mathsf{f}, \mathsf{f}')$ with

$$\varsigma(\mathsf{f}, \mathsf{f}') = \int_{\mathbb{R}^3} (fg' - gf')\, \mathrm{d}^3x, \qquad \mathsf{f}' = (f', g'), \tag{5}$$

so $\mathsf{W}(\mathsf{f})\mathsf{W}(\mathsf{f}') = e^{-\mathrm{i}\varsigma(\mathsf{f},\mathsf{f}')/2}\mathsf{W}(\mathsf{f} + \mathsf{f}')$, and $\mathsf{W}(\mathsf{f})$ and $\mathsf{W}(\mathsf{f}')$ commute when the supports of f and f' are disjoint ($\operatorname{supp}\mathsf{f} := \operatorname{supp} f \cup \operatorname{supp} g$), since then $\varsigma(\mathsf{f}, \mathsf{f}') = \varsigma(\mathsf{f}', \mathsf{f}) = 0$. For mass $\mu > 0$, the vacuum is the Gaussian (quasi-free) state [16]

$$\omega(\mathsf{W}(\mathsf{f})) = e^{-\mathsf{v}_\mu(\mathsf{f},\mathsf{f})/2},$$
$$\mathsf{v}_\mu(\mathsf{f}, \mathsf{f}') = \tfrac{1}{2}\big(\langle f, (\mu^2 - \nabla^2)^{-1/2} f'\rangle + \langle g, (\mu^2 - \nabla^2)^{1/2} g'\rangle\big), \tag{6}$$

with $\langle\cdot,\cdot\rangle$ the $L^2(\mathbb{R}^3)$ inner product; v_μ is the *covariance*, and $\mathsf{v}_\mu(\mathsf{f}, \mathsf{f}) = \omega(\Phi(\mathsf{f})^2)$. (The covariances of [17] are twice ours.) By Araki's time-slice theorem [18, 19],

$$\mathfrak{A}_\mu(D_R) = \{\mathsf{W}(f, g) : f, g \in C_c^\infty(B_R(0); \mathbb{R})\}'', \tag{7}$$

(the double prime is the double commutant, the smallest von Neumann algebra containing the set in braces), and likewise for the translated double cone with sources in $B_R(L\mathbf{e}_1)$. An *effect* is an operator $0 \leq E \leq I$, the quantum analogue of an event, with probability $\omega(E)$; a *projection* is an effect with $E^2 = E$; a self-adjoint contraction A has the two effects $(I \pm A)/2$, which are projections when $A^2 = I$, in which case we call A a *reflection*.

Lemma 1 (Dilation; cf. [20, Section 2])**.** $\beta_m(R, L) = \beta_{mR}(1, L/R)$ *for all $m, R > 0$ and $L > 2R$.*

Proof. The map $(f, g) \mapsto (\tilde f, \tilde g)$, $\tilde f(\mathbf{y}) = R^2 f(R\mathbf{y})$, $\tilde g(\mathbf{y}) = Rg(R\mathbf{y})$, carries sources in $B_R(0)$ and $B_R(L\mathbf{e}_1)$ to sources in $B_1(0)$ and $B_1(s\mathbf{e}_1)$, $s = L/R$, preserves (5), and carries the covariance of mass m to that of mass mR, since $\hat{\tilde f}(\mathbf{q}) = R^{-1}\hat f(\mathbf{q}/R)$, $\hat{\tilde g}(\mathbf{q}) = R^{-2}\hat g(\mathbf{q}/R)$ and $\sqrt{(mR)^2 + |\mathbf{q}|^2} = R\sqrt{m^2 + |\mathbf{q}/R|^2}$. It therefore induces a unitary between the two vacuum representations that fixes the vacuum and maps each local algebra of (7) onto its counterpart, so the two suprema (1) coincide. □

From now on $\mu > 0$ is arbitrary (the letter μ keeps this mass distinct from the physical m of Theorem 1), the balls are $B_1(0)$ and $B_1(s\mathbf{e}_1)$ with $s > 2$, and

$$\mathfrak{A} = \mathfrak{A}_\mu(D_1), \qquad \mathfrak{B} = \mathfrak{A}_\mu(D_1 + (0, s\mathbf{e}_1)).$$

Theorem 1 is the statement that $\beta(\omega; \mathfrak{A}, \mathfrak{B}) = 1$ whenever $s \geq C_*$.

One-particle vectors.—The one-particle space of the field of mass μ is $L^2(H_\mu^+, \mathrm{d}\mathfrak{m}_\mu)$, $H_\mu^+ = \{p : p_0 = \sqrt{\mu^2 + |\mathbf{p}|^2}\}$, $\mathrm{d}\mathfrak{m}_\mu = \mathrm{d}^3p/(2(2\pi)^3 p_0)$, and the one-particle vector of a source is

$$(\kappa\mathsf{f})(p) = \hat f(\mathbf{p}) + \mathrm{i}p_0 \hat g(\mathbf{p}). \tag{8}$$

By Plancherel's formula, for real sources,

$$\langle \kappa\mathsf{f}, \kappa\mathsf{f}'\rangle = \mathsf{v}_\mu(\mathsf{f}, \mathsf{f}') + \tfrac{\mathrm{i}}{2}\varsigma(\mathsf{f}, \mathsf{f}') \tag{9}$$

(inner products are antilinear in the first argument; on the left it is that of $L^2(H_\mu^+, \mathrm{d}\mathfrak{m}_\mu)$). For a nonempty bounded open $U \subset \mathbb{R}^3$, let $\mathcal{X}_{\mu,U}$ be the completion of the real smooth sources

in U in the norm $\mathsf{v}_\mu(\mathsf{f},\mathsf{f})^{1/2}$, and $\mathcal{H}_{\mu,U}$ its complexification, with inner product $\langle\cdot,\cdot\rangle_{\mu,U}$ extending v_μ sesquilinearly; by (9), κ extends to a bounded map $\kappa_U\colon \mathcal{H}_{\mu,U} \to L^2(H_\mu^+, \mathrm{d}\mathfrak{m}_\mu)$, isometric on real sources. Real-linear maps such as inclusions of source spaces are extended complex-linearly without change of notation.

Lemma 2 (Injectivity and density)**.** *κ_U is injective and has dense range.*

The proof (Appendix D) uses that the Fourier transforms of compactly supported sources are entire. The statement that κ_U has dense range is the one-particle form of the Reeh–Schlieder property; injectivity is what will give the strict uncertainty relation (12) below.

A. Modes and finite canonical systems

A *mode* is one canonical degree of freedom of the field in a region. This corresponds to a pair of sources f,f' there with $\varsigma(\mathsf{f},\mathsf{f}') = 1$, so that $[\Phi(\mathsf{f}),\Phi(\mathsf{f}')] = \mathrm{i}$ by (4). The local algebras of a free massive scalar carry no density matrices (Section IV), so every comparison in this paper is made on finitely many modes at a time: the degrees of freedom seen by a detector that couples to finitely many smeared fields $\Phi(\mathsf{f})$ in the region.

A *finite canonical system* is a finite collection of independent modes. Formally, it is a finite-dimensional real space F of smooth sources in a bounded open set, on which ς is nondegenerate: in a basis with $\varsigma(\mathsf{f}_k,\mathsf{f}_l) = (J_n)_{kl}$, $J_n = \bigoplus_{j=1}^{n} \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, the $2n$ operators $\Phi(\mathsf{f}_k)$ form n canonical pairs, $[\Phi(\mathsf{f}_{2j-1}),\Phi(\mathsf{f}_{2j})] = \mathrm{i}$ with all other commutators zero.

By the Stone–von Neumann theorem [16], the algebra generated by $\mathsf{W}(\mathsf{f})$, $\mathsf{f} \in F$, is a copy of the bounded operators on $L^2(\mathbb{R}^n)$—ordinary quantum mechanics of n degrees of freedom—possibly with multiplicity, and the vacuum restricted to it is given by a density matrix, since a vector state is normal. By (6) its characteristic function is Gaussian,

$$\omega\Big(\mathsf{W}\Big(\sum_k \xi_k \mathsf{f}_k\Big)\Big) = e^{-\xi^T V \xi/2} \tag{10}$$

for $\xi \in \mathbb{R}^{2n}$, and a density matrix is determined by its characteristic function [17]; so the restriction is the Gaussian state ρ_F with covariance matrix $V_{kl} = \mathsf{v}_\mu(\mathsf{f}_k,\mathsf{f}_l)$.

A finite collection of sources in an open set U need not form canonical pairs by itself: field sources $(f,0)$, for instance, all commute. Nevertheless, such a finite collection can always be completed into a finite canonical system supported inside U by adjoining finitely many further sources. The only obstruction to such a construction would be if there were a direction $\mathsf{f} = (f,g)$ in the span that commutes with the whole span, but we can always include the missing partner $(0,f)$ since $\varsigma(\mathsf{f},(0,f)) = \int f^2 > 0$ if $f \neq 0$ or $(-g,0)$ if $f = 0$ since $\varsigma(\mathsf{f},(-g,0)) = \int g^2 > 0$. Each such adjoined partner lowers the number of independent partnerless sources by one, so finitely many steps complete the pairs. Such a construction allows us to always complete two finite canonical systems in U into a common third system, which Lemma 3 below needs.

Given a Gaussian state on a given finite canonical system, Williamson's normal form [21] writes $V = S(\bigoplus_j \upsilon_j I_2)S^T$ with S a linear canonical transformation and I_2 the identity on the corresponding canonical pair. In these normal-mode coordinates ρ_F is a product of n oscillator states, the jth with covariance $\upsilon_j I_2$. Each normal mode is thermal: with N_j its number operator, the jth factor is proportional to $e^{-\beta_j N_j}$ for an inverse temperature $\beta_j \in (0,\infty]$, and the *symplectic eigenvalue* $\upsilon_j \geq \frac{1}{2}$ is related to the inverse temperature of the mode by

$$\upsilon_j = \tfrac{1}{2}\coth(\beta_j/2) = \bar{n}_j + \tfrac{1}{2}. \tag{11}$$

Deriving this is a standard exercise in the Heisenberg algebra together with equation (10). One can see that a pure mode has $\upsilon_j = \frac{1}{2}$ ($\beta_j = \infty$), while a mixed/thermal one has $\upsilon_j > \frac{1}{2}$. The vacuum restricted to a finite canonical system satisfies the strict uncertainty relation

$$V + \tfrac{\mathrm{i}}{2}J_n > 0, \quad \text{equivalently} \quad \upsilon_j > \tfrac{1}{2} \text{ for all } j, \tag{12}$$

because $V + \frac{\mathrm{i}}{2}J_n$ is the Gram matrix of the one-particle vectors of $\mathsf{f}_1,\dots,\mathsf{f}_{2n}$, which are linearly independent by Lemma 2 (see (9)). Hence ρ_F is *faithful*: it has no zero eigenvalues.

Lemma 3 (Passage to all local effects)**.** *Let h be continuous on $[0,1]^3$, and suppose that*

$$h\big(\omega(E),\omega(P),\omega(EP)\big) \leq 0$$

for every *finite canonical system F_A in $B_1(0)$, every finite canonical system F_B in $B_1(s\mathbf{e}_1)$, every effect E of the algebra of F_A and every effect P of the algebra of F_B. Then the inequality holds for all effects $E \in \mathfrak{A}$, $P \in \mathfrak{B}$. The same holds for a bound on $|\omega(AB) - \omega(A)\omega(B)|$ over self-adjoint contractions.*

Proof. The union of the algebras of all finite canonical systems in $B_1(0)$ is a $*$-algebra (two systems lie in a third system) containing the generators of $\mathfrak{A}$, so it is strongly dense in $\mathfrak{A}$. By Kaplansky's density theorem [22, Theorem 2.4.16], every self-adjoint contraction of $\mathfrak{A}$ is a strong limit of self-adjoint contractions of the union; applied to $2E - I$ this gives effects $E_\alpha \to E$ strongly, each belonging to the algebra of some finite canonical system, and likewise $P_\beta \to P$ in $\mathfrak{B}$. Products of bounded strongly convergent nets converge strongly, and the vacuum, a vector state, is strongly continuous (as is any normal state, on bounded sets), so $\big(\omega(E_\alpha),\omega(P_\beta),\omega(E_\alpha P_\beta)\big) \to \big(\omega(E),\omega(P),\omega(EP)\big)$, and by the continuity of h the closed condition $h \leq 0$ passes to the limit. □

III. A PROBABILITY CRITERION FOR THE CLASSICAL BOUND

In this section ω is any state on the algebra generated by two commuting von Neumann algebras $\mathfrak{A},\mathfrak{B}$. For $r > 1$, the *event*

comparison constants are

$$K_{r,+} = \sup_{E,P} \frac{\omega(EP)^r}{(\omega(E)\omega(P))^{r-1}}, \quad K_{r,-} = \sup_{E,P} \frac{(\omega(E)\omega(P))^r}{\omega(EP)^{r-1}}, \tag{13}$$

over projections $E \in \mathfrak{A}$, $P \in \mathfrak{B}$ with $\omega(E)\omega(P) > 0$; a positive numerator over a zero denominator counts as $+\infty$. Both constants are at least 1 ($E = P = I$). For $r_+ > 1$ and $r_- > 1$, taking roots in (13) gives, for all projections,

$$\begin{aligned} \omega(EP) &\le K_{r_+,+}^{1/r_+} \big(\omega(E)\omega(P)\big)^{1-1/r_+}, \\ \omega(EP) &\ge K_{r_-,-}^{-1/(r_--1)} \big(\omega(E)\omega(P)\big)^{r_-/(r_--1)}; \end{aligned} \tag{14}$$

if $\omega(E)\omega(P) = 0$ then $\omega(EP) = 0$ since $EP \le E, P$. Thus a finite $K_{r,+}$ says that a joint event is never much more probable than a power of the product of its marginals, and a finite $K_{r,-}$ says it is never much less probable.

Theorem 2 (Two event bounds and clustering)**.** *Let $r_+ > 2$, $1 < r_- < 2$, and*

$$\begin{aligned} b = 1 - \frac{1}{r_+}, \quad a = 1 - \frac{1}{r_-}, \quad \eta = 1 - \frac{1-b}{ab^2}, \\ \nu_+ = \frac{1+b}{ar_+b^2}, \quad \nu_- = \frac{1-a}{a}. \end{aligned} \tag{15}$$

Suppose $\eta > 0$, $K_{r_+,+}, K_{r_-,-} < \infty$, and, for some $q_{\mathrm{cl}} \ge 0$,

$$|\omega(AB) - \omega(A)\omega(B)| \le q_{\mathrm{cl}} \tag{16}$$

for all self-adjoint contractions $A \in \mathfrak{A}$, $B \in \mathfrak{B}$. If

$$q_{\mathrm{cl}}^{\eta} K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} \le 2^{-2/(ab)}, \tag{17}$$

then $\beta(\omega; \mathfrak{A}, \mathfrak{B}) = 1$.

The proof is in Appendix A. In outline: a self-adjoint contraction is an average of reflections and the CHSH expression is affine in each argument, so a violation by contractions implies a violation by four reflections A_0, A_1, B_0, B_1. Their spectral projections $E_i^{\pm} = (I \pm A_i)/2$ and $P_j^{\pm} = (I \pm B_j)/2$ define sixteen joint probabilities $\omega(E_i^{\pm} P_j^{\pm})$, and a violation forces the least likely of the eight that enter the CHSH value with a positive sign—by the symmetries of the expression, this cell can be taken to be $E_0^- P_0^-$—to satisfy two incompatible bounds: the event bounds (14) give

$$\omega(E_0^- P_0^-)^{\eta} K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} > 2^{-2/(ab)}, \tag{18}$$

while (16) forces $\omega(E_0^- P_0^-) < q_{\mathrm{cl}}$, and together these contradict (17).

What remains.—Theorem 2 reduces Theorem 1 to verifying its hypotheses for the vacuum and the two algebras $\mathfrak{A}, \mathfrak{B}$ at large separation s: bounds on $K_{r_+,+}, K_{r_-,-}$ tending to 1, and a clustering constant q_{cl} tending to 0, fast enough that (17) holds for all $s \ge C_*$. Both come from comparing the vacuum with the product of its marginals. The comparison degenerates on modes that are nearly pure in the vacuum; it is rescued by the monogamy stated in Section I—purity of a mode in one ball suppresses its correlation with the other ball at a faster rate. Section IV bounds the quotients in (13), for the vacuum restricted to finitely many modes, by Rényi divergences between the restricted vacuum and the product of its marginals, and bounds q_{cl} by their trace distance; Lemma 3 then removes the restriction to finitely many modes. The divergences are bounded in turn by covariance differences (Theorem 3), which Section V bounds by negative powers of s, uniformly in the mass. Section VI fixes $r_\pm$ and assembles (17) with q_{cl} proportional to $1/s$.

IV. GAUSSIAN COMPARISON AND THE WEIGHT OPERATOR

This section carries out the first step of the programme at the end of Section III: for the vacuum restricted to finitely many modes in the two balls, the quotients in (13) are bounded by Rényi divergences between the restricted vacuum ρ_F and the product σ_F of its marginals. Furthermore, q_{cl} is bounded by their trace distance. Theorem 3 then bounds the divergences by covariance differences as in (24). The section then confronts the main difficulty of the argument, uniformity over the choice of modes, and resolves it by the monogamy mechanism of Section I.

A divergence with data processing.—For faithful density operators ρ, σ and $r > 1$, let

$$\widetilde{Q}_r(\rho\|\sigma) = \mathrm{Tr}\big(\sigma^{-\tau/2} \rho\, \sigma^{-\tau/2}\big)^r, \qquad \tau = \frac{r-1}{r}, \tag{19}$$

with the value $+\infty$ if the operator inside has infinite r-norm. Its logarithm divided by $r - 1$ is the sandwiched Rényi divergence [23, 24], a one-parameter family of measures of how distinguishable two states are; $\widetilde{Q}_r = 1$ if $\rho = \sigma$ and $\widetilde{Q}_r \ge 1$ in general. Its basic property is data processing: it cannot increase under any physical operation [25, 26]. We use two consequences. Here $\rho(M) = \mathrm{Tr}(\rho M)$, and $\|T\|_p = (\mathrm{Tr}|T|^p)^{1/p}$ are the Schatten norms; $\|T\|_1$, the *trace norm*, is the sum of the singular values.

Lemma 4 (Effects and trace distance; cf. [23, 25–27])**.** *Let ρ, σ be faithful density operators on one Hilbert space, with $\widetilde{Q}_r(\rho\|\sigma) < \infty$. Then every effect M on that space satisfies*

$$\rho(M)^r \le \widetilde{Q}_r(\rho\|\sigma)\, \sigma(M)^{r-1}, \tag{20}$$

and, for $r \ge 2$,

$$\|\rho - \sigma\|_1 \le \sqrt{\widetilde{Q}_r(\rho\|\sigma)^{1/(r-1)} - 1}. \tag{21}$$

The proof is in Appendix B. A word on where density matrices live: the restrictions of the vacuum to the local algebras $\mathfrak{A}, \mathfrak{B}$ are normal states, but these algebras are von Neumann algebras of type III, without a trace, so the restricted states have no intrinsic density matrix there. Density matrices exist on the finite canonical systems of Section II, which are ordinary quantum mechanics by the Stone–von Neumann theorem; the

density matrices used below represent the vacuum restricted to such systems, and every trace below is taken there. In the end, the resulting inequalities will be between the numbers $\omega(E)$, $\omega(P)$, $\omega(EP)$ with constants independent of the finite system, and so the limits can be uniformly taken to $\mathfrak{A}, \mathfrak{B}$, by Lemma 3.

Returning to the vacuum ω, let F_A, F_B be finite canonical systems (finitely many modes) in $B_1(0)$ and $B_1(s\mathbf{e}_1)$. Since ς vanishes between them, $F = F_A \oplus F_B$ is a finite canonical system whose two halves generate commuting algebras. Let ρ_F be the vacuum on F and

$$\sigma_F = \rho_{F_A} \otimes \rho_{F_B},$$

the product of the vacuum restrictions to the two halves (the marginals of ρ_F), both faithful Gaussian states by (12). For effects E of F_A and P of F_B, EP is an effect with $\rho_F(EP) = \omega(EP)$ and $\sigma_F(EP) = \omega(E)\omega(P)$. Applying (20) to (ρ_F, σ_F, r_+) and to (σ_F, ρ_F, r_-) with $M = EP$,

$$\begin{aligned} \omega(EP)^{r_+} &\le \widetilde{Q}_{r_+}(\rho_F\|\sigma_F)\,\big(\omega(E)\omega(P)\big)^{r_+-1}, \\ \big(\omega(E)\omega(P)\big)^{r_-} &\le \widetilde{Q}_{r_-}(\sigma_F\|\rho_F)\,\omega(EP)^{r_--1}. \end{aligned} \tag{22}$$

These are exactly the quotients of (13). The task is to bound $\log \widetilde{Q}_{r_+}(\rho_F\|\sigma_F)$ and $\log \widetilde{Q}_{r_-}(\sigma_F\|\rho_F)$ by quantities that vanish as $s \to \infty$, uniformly in μ and in F_A, F_B.

Thermal powers.—If ρ_V is a faithful Gaussian state with covariance V and $0 < \tau < 1$, then $\rho_V^\tau/\operatorname{Tr}\rho_V^\tau$ is Gaussian with covariance $V^{[\tau]}$, which has the same normal modes as V and the symplectic eigenvalues [17, Proposition 1]

$$\upsilon_j^{[\tau]} = \tfrac{1}{2}\coth\Big(\tau\,\operatorname{artanh}\frac{1}{2\upsilon_j}\Big), \tag{23}$$

the τth power of a thermal state being the thermal state at τ times the inverse temperature. The *thermal increment* $\mathsf{H}_\tau(V) = V^{[\tau]} - V$ is positive definite, since coth decreases.

Theorem 3 (Gaussian Rényi comparison)**.** *Let ρ, σ be centred (zero-mean) faithful Gaussian states of the same n modes, with covariance matrices V_ρ, V_σ. For $r > 1$ let*

$$\begin{aligned} \tau = \frac{r-1}{r}, \qquad k_r = \max\{2, \tau^{-1}\}, \\ \mathsf{Z} = \mathsf{H}_\tau(V_\sigma)^{-1/2}(V_\rho - V_\sigma)\mathsf{H}_\tau(V_\sigma)^{-1/2}. \end{aligned}$$

If $\|\mathsf{Z}\| \le (4k_r)^{-1}$, then

$$\log \widetilde{Q}_r(\rho\|\sigma) \le 2rk_r\|\mathsf{Z}\|_1. \tag{24}$$

The proof is in Appendix B. Neither n nor the symplectic eigenvalues appear: the covariance difference enters only relative to the thermal increment of σ.

The obstruction: nearly pure modes.—Theorem 3 and (22) cannot be combined as they stand to prove the main result. Lemma 3 demands constants independent of the chosen modes F, so $\|\mathsf{Z}\|_1$ must first be bounded uniformly in F—and Z is the covariance difference *divided by* the thermal increment of σ_F. Consider a single normal mode of σ_F, thermal at inverse temperature β_j as in (11). By (23) and (11), the thermal increment on the mode takes the form

$$\begin{aligned} \upsilon_j^{[\tau]} - \upsilon_j &= \tfrac{1}{2}\coth(\tau\beta_j/2) - \tfrac{1}{2}\coth(\beta_j/2) \\ &\sim e^{-\tau\beta_j}, \qquad \beta_j \to \infty, \end{aligned}$$

and so on that mode each side of Z divides the covariance difference by a quantity of order $e^{-\tau\beta_j/2}$. Nothing stops F from containing modes with β_j as large as one likes (physically, vacuum entanglement concentrates near the boundary of a region, and modes supported well inside a ball are nearly pure), so no bound on the covariance difference alone controls $\|\mathsf{Z}\|_1$ uniformly in F.

The rescue: purity is monogamous.—So what must be shown is that the covariance difference itself decays on nearly pure modes, at least as fast as $e^{-\tau\beta_j/2}$. Since σ_F is the product of the marginals of ρ_F, the diagonal blocks of $V_\rho - V_\sigma$ cancel: what remains is the covariance between the two balls. And the vacuum is a pure state: across the split between the mode and the environment, the vacuum vector has a Schmidt decomposition with the thermal weights of (11),

$$\Omega = \sum_{n\ge 0}\sqrt{p_n}\,|n\rangle \otimes |\widetilde{n}\rangle, \qquad p_n = (1 - e^{-\beta_j})\,e^{-\beta_j n},$$

with $|n\rangle$ the n-quanta state of the mode and $|\widetilde{n}\rangle$ orthonormal vectors of the environment, which here means everything other than the mode. Let q be the position of the mode, which connects $|n\rangle$ only to $|n \pm 1\rangle$, with $\langle n+1|q|n\rangle = \sqrt{(n+1)/2}$, and let f' be any source in the other ball, so that the smeared field $\Phi(\mathsf{f}')$ acts on the environment factor alone, with $\omega(\Phi(\mathsf{f}')) = 0$. The covariance is

$$\begin{aligned} \omega(q\,\Phi(\mathsf{f}')) &= \sum_n \sqrt{p_n p_{n+1}}\,\sqrt{2(n+1)}\,\operatorname{Re}\widetilde{\langle n+1|\Phi(\mathsf{f}')|\widetilde{n}\rangle} \\ &= e^{-\beta_j/2}\sum_n p_n\sqrt{2(n+1)}\,\operatorname{Re}\widetilde{\langle n+1|\Phi(\mathsf{f}')|\widetilde{n}\rangle}, \end{aligned}$$

because $p_{n+1} = e^{-\beta_j}p_n$: each term pays $e^{-\beta_j/2}$ for moving one step along the Schmidt sum. The Cauchy–Schwarz inequality and $\sum_n p_n\langle\widetilde{n}|\Phi(\mathsf{f}')^2|\widetilde{n}\rangle = \omega(\Phi(\mathsf{f}')^2) = \mathsf{v}_\mu(\mathsf{f}',\mathsf{f}')$ give

$$\begin{aligned} |\omega(q\,\Phi(\mathsf{f}'))| &\le e^{-\beta_j/2}\sqrt{2(\bar{n}_j+1)}\;\mathsf{v}_\mu(\mathsf{f}',\mathsf{f}')^{1/2} \\ &= \sqrt{2\bar{n}_j}\;\mathsf{v}_\mu(\mathsf{f}',\mathsf{f}')^{1/2}, \end{aligned}$$

where $\bar{n}_j = \sum_n np_n = 1/(e^{\beta_j} - 1)$ is the mean number of quanta in the mode, as in (11), so that $e^{-\beta_j}(\bar{n}_j + 1) = \bar{n}_j$; likewise for the momentum. A pure mode ($\bar{n}_j = 0$) is therefore correlated with nothing—a pure state with a pure marginal factorizes—and every covariance of a nearly pure mode with its environment is suppressed by $\sqrt{2\bar{n}_j} \sim \sqrt{2}\,e^{-\beta_j/2}$. Since $\tau < 1$, the covariance difference decays faster than the conjugation by $\mathsf{H}_\tau(V_\sigma)^{-1/2}$ in Z grows.

Showing this rigorously is the plan for the rest of the proof. The operator R, constructed next, acts on the full source space of a ball and on mode j as a multiple of $e^{-\beta_j/2}$: it is the per-mode factor of the covariance bound above, made independent

of F. Dividing this factor out of the covariance difference on each side defines, schematically,

$$Y = \mathsf{R}^{-\tau}(V_\rho - V_\sigma)\mathsf{R}^{-\tau}, \quad \text{i.e.} \quad V_\rho - V_\sigma = \mathsf{R}^{\tau} Y \mathsf{R}^{\tau}.$$

Since R has eigenvalues accumulating at 0, $\mathsf{R}^{-\tau}$ is unbounded and Y could a priori be infinite. Section V (Theorems 4 and 5) shows that it is not: Y is trace class, with $\|Y\|_1$ decaying in s and independent of F—the surplus $e^{-(1-\tau)\beta_j/2}$ per mode is what pays for this. In Z, each factor R^{τ} then cancels one factor $\mathsf{H}_\tau(V_\sigma)^{-1/2}$: Lemma 5 below shows that the two vanish at the same rate, Lemma 6 shows that restricting to F does not spoil this, and the resulting bounds (31) and (32) on $\|\mathsf{Z}\|_1$ do not depend on F.

The weight operator.—We now construct R, first on the n modes of F. Choose coordinates in which $V_\sigma = I$, and let J be the real antisymmetric matrix of commutators in these coordinates, $[\Phi(\mathsf{f}_k), \Phi(\mathsf{f}_l)] = \mathrm{i}\mathsf{J}_{kl}$, and let

$$G = I + \tfrac{\mathrm{i}}{2}\mathsf{J}, \qquad \overline{G} = I - \tfrac{\mathrm{i}}{2}\mathsf{J}.$$

In both applications below, σ is the vacuum restricted to the modes in question (to the modes of one ball, or to all of F), so $V_\sigma = I$ means $\mathsf{v}_\mu(\mathsf{f}_k, \mathsf{f}_l) = \delta_{kl}$, and by (9) $G_{kl} = \langle \kappa \mathsf{f}_k, \kappa \mathsf{f}_l \rangle$ is the matrix of inner products of the one-particle vectors. In normal-mode coordinates rescaled to $V_\sigma = I$, mode j has covariance I_2 and commutator block J_1/υ_j, so G and $\overline{G}$ have eigenvalues $1 \pm d_j$, with

$$d_j = \frac{1}{2\upsilon_j} = \tanh(\beta_j/2) \in (0,1)$$

by (11) and (12). Write d for the matrix $|\tfrac{\mathrm{i}}{2}\mathsf{J}|$, with eigenvalues d_j. The *weight operator* is

$$\mathsf{R} = (G\overline{G})^{1/2} = (I - d^2)^{1/2}, \tag{25}$$

which acts on mode j as $\operatorname{sech}(\beta_j/2) \sim 2e^{-\beta_j/2}$: it vanishes exactly on pure modes, at the rate of the covariance bound above, and tends to 1 on strongly mixed ones. In the same coordinates the thermal increment (23) is

$$\mathsf{H}_\tau(I) = d \coth(\tau \operatorname{artanh} d) - I, \tag{26}$$

which acts on mode j as $\tanh(\beta_j/2)\coth(\tau\beta_j/2) - 1 \sim 2e^{-\tau\beta_j}$, the same rate as $\mathsf{R}^{2\tau}$.

Nothing in this construction needs F. On the full source space $\mathcal{H}_{\mu,U}$ of an open set U, with inner product v_μ, define J by $\mathsf{v}_\mu(\mathsf{f}, \mathsf{J}\mathsf{f}') = \varsigma(\mathsf{f}, \mathsf{f}')$; then, again by (9),

$$\begin{aligned} G_U = \kappa_U^* \kappa_U = I + \tfrac{\mathrm{i}}{2}\mathsf{J}, \\ \overline{G}_U = 2I - G_U, \qquad \mathsf{R}_U = (G_U \overline{G}_U)^{1/2}. \end{aligned} \tag{27}$$

By Lemma 2, G_U and $\overline{G}_U$ are injective with spectrum in $[0,2]$, so $0 < \mathsf{R}_U \le I$; now the spectrum of R_U may accumulate at 0. A finite canonical system $F \subset U$ sits inside $\mathcal{H}_{\mu,U}$ through its inclusion ι, and the matrix of an operator X on $\mathcal{H}_{\mu,U}$ between vectors of F is $\iota^* X \iota$. The next two lemmas, proved in Appendix C, compare the thermal increment with R, and compare R for F with $\iota^* \mathsf{R}_U \iota$.

Lemma 5 (Thermal comparison)**.** *On any source space, finite or full, and for* $0 < \tau < 1$*,*

$$\mathsf{H}_\tau(I) \ge c_\tau \mathsf{R}^{2\tau}, \qquad c_\tau = \begin{cases} 2^{1-2\tau}, & 0 < \tau \le \frac{1}{2}, \\ (1-\tau)/\tau, & \frac{1}{2} \le \tau < 1, \end{cases} \tag{28}$$

where on the full source space $\mathsf{H}_\tau(I)$ *is defined by* (26) *in the functional calculus of* d*.*

Lemma 6 (Compression)**.** *Let* ι *be the inclusion of a finite canonical system* F *into a full source space—isometric for the covariance norm and preserving* ς*, so that* $\iota^*\mathsf{J}\iota$ *is the commutator form of* F*—and let* $G_F = \iota^* G \iota$*,* $\overline{G}_F = \iota^* \overline{G} \iota$*,* $\mathsf{R}_F = (G_F \overline{G}_F)^{1/2}$ *be the operators* (27) *of* F *itself. For* $0 < a \le \frac{1}{2}$*,*

$$\mathsf{R}_F^{2a} \ge \iota^* \mathsf{R}^{2a} \iota, \tag{29}$$

and for an integer $r \ge 2$ *and* $b = (r-1)/r$*,*

$$\mathsf{H}_b(I_F) \ge \iota^* \mathsf{H}_b(I) \iota \ge (r-1)^{-1} \iota^* \mathsf{R}^{2b} \iota. \tag{30}$$

Together, the two lemmas give the cancellation announced above. For $0 < a \le \frac{1}{2}$, (28) on F and (29) give $\mathsf{H}_a(I_F) \ge 2^{1-2a} \iota^* \mathsf{R}^{2a} \iota$, that is, $\|\mathsf{R}^a \iota \, \mathsf{H}_a(I_F)^{-1/2}\|^2 \le 2^{2a-1}$. Since $\|ABC\|_1 \le \|A\| \|B\|_1 \|C\|$, every trace-class Y satisfies

$$\begin{aligned} \big\| \mathsf{H}_a(I_F)^{-1/2} \iota^* \mathsf{R}^a Y \mathsf{R}^a \iota \, \mathsf{H}_a(I_F)^{-1/2} \big\|_1 \\ \le 2^{2a-1} \|Y\|_1. \end{aligned} \tag{31}$$

In the same way, for the two balls, with weight operators $\mathsf{R}_A = \mathsf{R}_{B_1(0)}$ and $\mathsf{R}_B = \mathsf{R}_{B_1(s\mathbf{e}_1)}$ and inclusions $\iota_A \colon F_A \to \mathcal{X}_{\mu,B_1(0)}$ and $\iota_B \colon F_B \to \mathcal{X}_{\mu,B_1(s\mathbf{e}_1)}$, (30) gives, for an integer $r \ge 2$, $b = (r-1)/r$ and every trace-class X,

$$\begin{aligned} \big\| \mathsf{H}_b(I_{F_A})^{-1/2} \iota_A^* \mathsf{R}_A^b X \mathsf{R}_B^b \iota_B \, \mathsf{H}_b(I_{F_B})^{-1/2} \big\|_1 \\ \le (r-1) \|X\|_1. \end{aligned} \tag{32}$$

V. THE TWO FIELD ESTIMATES

This section proves the factorization $V_\rho - V_\sigma = \mathsf{R}^{\tau} Y \mathsf{R}^{\tau}$ of Section IV, with $\|Y\|_1$ bounded by a negative power of s uniformly in μ. It is needed twice, because (22) uses two divergences: the *forward* comparison $\widetilde{Q}_{r_+}(\rho_F \| \sigma_F)$, which conjugates by the thermal increment of σ_F, and the *reverse* comparison $\widetilde{Q}_{r_-}(\sigma_F \| \rho_F)$, which conjugates by that of ρ_F. For each there is one operator on full source spaces whose matrix on F is the covariance difference.

For the forward comparison it is the *cross-covariance operator* $\mathsf{C}_{\mu,s} \colon \mathcal{H}_{\mu,B_1(s\mathbf{e}_1)} \to \mathcal{H}_{\mu,B_1(0)}$, defined by

$$\langle \mathsf{f}, \mathsf{C}_{\mu,s} \mathsf{f}' \rangle_{\mu,B_1(0)} = \mathsf{v}_\mu(\mathsf{f}, \mathsf{f}'). \tag{33}$$

For the reverse comparison, let $U_s = B_1(0) \cup B_1(s\mathbf{e}_1)$ and split each source on U_s as $\mathsf{f} = \mathsf{f}_A + \mathsf{f}_B$ into its parts in the two balls. The *product-minus-joint operator* $\mathsf{D}_{\mu,s}$ on $\mathcal{H}_{\mu,U_s}$, which

carries the joint covariance v_μ as inner product, is defined by deleting the cross terms:

$$\begin{aligned}\langle \mathsf{f}, \mathsf{D}_{\mu,s}\mathsf{f}'\rangle_{\mu,U_s} &= \mathsf{v}_\mu(\mathsf{f}_A,\mathsf{f}'_A) + \mathsf{v}_\mu(\mathsf{f}_B,\mathsf{f}'_B) - \mathsf{v}_\mu(\mathsf{f},\mathsf{f}')\\ &= -\mathsf{v}_\mu(\mathsf{f}_A,\mathsf{f}'_B) - \mathsf{v}_\mu(\mathsf{f}_B,\mathsf{f}'_A).\end{aligned} \tag{34}$$

Section VI checks that $\iota_A^*\mathsf{C}_{\mu,s}\iota_B$ is the off-diagonal block of $V_{\rho_F} - V_{\sigma_F}$ in coordinates with $V_{\sigma_F} = I$, and that $\iota_F^*\mathsf{D}_{\mu,s}\iota_F = V_{\sigma_F} - V_{\rho_F}$ in coordinates with $V_{\rho_F} = I$, where ι_F is the inclusion of F into $\mathcal{H}_{\mu,U_s}$. The trace-class factor called Y in Section IV is $X_{b,\mu,s}$ below for the forward comparison and $Y_{\mu,s}$ for the reverse one.

Theorem 4 (Forward estimate)**.** *For each $\frac{1}{2} \le b < 1$ there are finite S_b, F_b, independent of μ, such that for all $\mu > 0$ and $s \ge S_b$ there is a trace-class $X_{b,\mu,s}$ with*

$$\mathsf{C}_{\mu,s} = \mathsf{R}_A^b X_{b,\mu,s}\mathsf{R}_B^b, \qquad \|X_{b,\mu,s}\|_1 \le F_b s^{-2}. \tag{35}$$

Theorem 5 (Reverse estimate)**.** *Let $a = 2^{-18}$, $\gamma = 2(1-2^{-17})$, and let R_{U_s} be the weight operator (27) of U_s. There are finite $S_{\mathrm{rev}}, \mathcal{J}_0$, independent of μ, such that for all $\mu > 0$ and $s \ge S_{\mathrm{rev}}$ there is a self-adjoint trace-class $Y_{\mu,s}$ with*

$$\mathsf{D}_{\mu,s} = \mathsf{R}_{U_s}^a Y_{\mu,s}\mathsf{R}_{U_s}^a, \quad \|Y_{\mu,s}\|_1 \le \mathcal{J}_0 s^{-\gamma}. \tag{36}$$

As in Section IV, the content of each theorem is that the middle factor exists and is trace class, although R_A^{-b}, R_B^{-b} and $\mathsf{R}_{U_s}^{-a}$ are unbounded. The decay is uniform in μ, including $\mu s \to 0$; since at $\mu = 0$ vacuum correlations decay only as a power of the distance, it can only be polynomial. The proofs are in Appendix E. They rest on the following bound, which controls the one-particle vector $\kappa\mathsf{f}$ of a source in a ball, continued to complex momenta, by the weighted norm $\|\mathsf{R}_U^b\mathsf{f}\|$.

Theorem 6 (Weighted complex-shell evaluation)**.** *For a source $\mathsf{f} = (f, g)$ let $\Psi_\mathsf{f}(z) = \hat{f}(\mathbf{z}) + \mathrm{i}z_0\hat{g}(\mathbf{z})$, $z \in \mathbb{C}^4$, an entire function that equals $\kappa\mathsf{f}$ on H_μ^+, and let $H_\mu^{\mathbb{C}} = \{z \in \mathbb{C}^4 : z\cdot z = -\mu^2\}$ be the complex mass shell, where $z\cdot z = -z_0^2 + \mathbf{z}\cdot\mathbf{z}$ is the Minkowski product extended bilinearly, without complex conjugation. For every $0 < c < \frac{1}{2}$ there are finite $C_c, \lambda_c \ge 1$ such that for all $\mu, T > 0$, nonempty bounded open $U \subset B_T(0)$, $\mathsf{f} \in \mathcal{H}_{\mu,U}$ and every energy scale $\mathcal{E} \ge \mu$,*

$$|\Psi_\mathsf{f}(z)|^2 \le C_c\mathcal{E}^{-2}e^{2\lambda_c T\mathcal{E}}\|G_U^{1/2-c}\overline{G}_U^{\,c}\mathsf{f}\|^2 \tag{37}$$

for all $z \in H_\mu^{\mathbb{C}}$ with $|z|_{\mathrm{E}} \le 2\mathcal{E}$ (Euclidean norm on $\mathbb{C}^4$); $\lambda_c = 2048/\sin(2\pi c)$ is admissible.

The right-hand side is a weighted norm of f. Indeed, let $\frac{1}{2} \le b < 1$. Since $\mathsf{R}_U^b = G_U^{b/2}\overline{G}_U^{\,b/2}$ and $G_U + \overline{G}_U = 2I$,

$$G_U^{1/2-c}\overline{G}_U^{\,c} = \begin{cases}\mathsf{R}_U^b\,\overline{G}_U^{(1-2b)/2}, & c = \frac{1-b}{2},\\ \mathsf{R}_U^b\,G_U^{(1-2b)/2}, & c = \frac{b}{2},\end{cases}$$

and the second factor is at most 1 on the part of the spectrum where $G_U \le 1$ in the first case and where $G_U \ge 1$ in the second. Using each choice on its part, for the same z,

$$|\Psi_\mathsf{f}(z)| \le \big(C^{1/2}_{(1-b)/2} + C^{1/2}_{b/2}\big)\mathcal{E}^{-1}e^{\lambda_{b/2}T\mathcal{E}}\|\mathsf{R}_U^b\mathsf{f}\|, \tag{38}$$

with $\lambda_{(1-b)/2} = \lambda_{b/2} = 2048/\sin(\pi b)$.

The bound (37) comes from the Bisognano–Wichmann theorem, which needs the following definitions. A closed real subspace $\mathcal{K}$ of a complex Hilbert space is *standard* if $\mathcal{K} \cap \mathrm{i}\mathcal{K} = \{0\}$ and $\mathcal{K} + \mathrm{i}\mathcal{K}$ is dense; the antilinear map $\mathsf{s}_\mathcal{K}(u + \mathrm{i}v) = u - \mathrm{i}v$, $u, v \in \mathcal{K}$, is then well defined and closable, and $\Delta_\mathcal{K} = \mathsf{s}_\mathcal{K}^*\mathsf{s}_\mathcal{K}$ is a positive operator, the *modular operator* of $\mathcal{K}$ [28]. For a region, take the closed real span of the one-particle vectors of the real sources it contains: $\mathcal{K}_U$ for the ball U, standard by Lemma 2, and $\mathcal{K}_{\mathcal{W}_\ell}$ for the wedge $\mathcal{W}_\ell = \{x^1 < \ell - |x^0|\}$, with sources in the base $\{x^1 < \ell\}$. The wedge subspace $\mathcal{K}_{\mathcal{W}_\ell}$ is standard too: it contains the ball's $\mathcal{K}_U$, so $\mathcal{K}_{\mathcal{W}_\ell} + \mathrm{i}\mathcal{K}_{\mathcal{W}_\ell}$ is dense; and if ξ and $-\mathrm{i}\xi$ both lie in $\mathcal{K}_{\mathcal{W}_\ell}$, then for every source f' in a ball U' in the opposite half-space $\{x^1 > \ell\}$ the numbers $\langle\xi, \kappa_{U'}\mathsf{f}'\rangle$ and $\mathrm{i}\langle\xi, \kappa_{U'}\mathsf{f}'\rangle$ are both real, because ς vanishes between the half-spaces (9); so $\xi \perp \kappa_{U'}\mathsf{f}'$, these vectors are dense by Lemma 2, and $\xi = 0$.

The Bisognano–Wichmann theorem [29], in its one-particle form [30], says that $\Delta_{\mathcal{W}_\ell}^{\mathrm{i}y}$ is the boost of rapidity $2\pi y$ that fixes the edge $\{x^0 = 0, x^1 = \ell\}$ of the wedge, that is, the boost along the first axis conjugated by the translation by $\ell\mathbf{e}_1$. Continued to imaginary y, $\Delta_{\mathcal{W}_\ell}^c$ for $0 < c < \frac{1}{2}$ is a boost by the imaginary rapidity $\mathrm{i}\varphi$, $\varphi = 2\pi c$, and acts on $\kappa_U\mathsf{f}$ by evaluating Ψ_f at complex momenta; the translation contributes the factor $|e^{\mathrm{i}\ell(p_1\cos\varphi + \mathrm{i}p_0\sin\varphi)}|^2 = e^{-2\ell p_0\sin\varphi}$:

$$\|\Delta_{\mathcal{W}_\ell}^c\kappa_U\mathsf{f}\|^2 = \int_{H_\mu^+} e^{-2\ell p_0\sin\varphi}|\Psi_\mathsf{f}(b_\varphi(p))|^2\,\mathrm{d}\mathfrak{m}_\mu(p), \tag{39}$$

with

$$b_\varphi(p) = (p_0\cos\varphi + \mathrm{i}p_1\sin\varphi,\ p_1\cos\varphi + \mathrm{i}p_0\sin\varphi,\ p_2,\ p_3).$$

To bring in R_U, let $U \subset B_T(0)$ with $T \le \ell$, so that $D_T \subset \mathcal{W}_\ell$ and $\mathcal{K}_U \subset \mathcal{K}_{\mathcal{W}_\ell}$. Then $\mathsf{s}_{\mathcal{K}_U}$ is a restriction of $\mathsf{s}_{\mathcal{K}_{\mathcal{W}_\ell}}$, so $\|\Delta_{\mathcal{W}_\ell}^{1/2}\xi\| = \|\Delta_{\mathcal{K}_U}^{1/2}\xi\|$ on the domain of $\mathsf{s}_{\mathcal{K}_U}$, and $\Delta_{\mathcal{W}_\ell} \le \Delta_{\mathcal{K}_U}$ as quadratic forms. Because $t \mapsto t^{2c}$ is operator monotone for $0 < 2c < 1$ (Löwner–Heinz [31]), $\Delta_{\mathcal{W}_\ell}^{2c} \le \Delta_{\mathcal{K}_U}^{2c}$. Finally, with the polar decomposition $\kappa_U = \mathcal{U}_U G_U^{1/2}$, one has $\Delta_{\mathcal{K}_U} = \mathcal{U}_U\overline{G}_U G_U^{-1}\mathcal{U}_U^*$ (Appendix D), so

$$\|\Delta_{\mathcal{W}_\ell}^c\kappa_U\mathsf{f}\| \le \|G_U^{1/2-c}\overline{G}_U^{\,c}\mathsf{f}\|. \tag{40}$$

Together, (39) and (40) bound an integral of $|\Psi_\mathsf{f}|^2$ over the complex momenta $b_\varphi(p)$, $p \in H_\mu^+$, by $\|G_U^{1/2-c}\overline{G}_U^{\,c}\mathsf{f}\|^2$. Appendix D turns this into the pointwise bound (37) in three steps. (i) Repeating the bound for a compact family of Lorentz transforms of the wedge and averaging bounds the integral of $|\Psi_\mathsf{f}|^2$ over an open patch of $H_\mu^{\mathbb{C}}$. (ii) Ψ_f is holomorphic, so $|\Psi_\mathsf{f}|^2$ at a point is at most its average over a small neighbourhood in $H_\mu^{\mathbb{C}}$; this bounds $|\Psi_\mathsf{f}|^2$ on the patch. (iii) For the same reason, $|\Psi_\mathsf{f}|^2$ at the centre of a complex disc lying in $H_\mu^{\mathbb{C}}$ is at most its average over the boundary circle; every $z \in H_\mu^{\mathbb{C}}$ with $|z|_{\mathrm{E}} \le 2\mathcal{E}$ is the centre of such a disc whose boundary lies in the patch.

Outline of the proofs of Theorems 4 and 5.—At the two shell points $z_\pm = (\pm\sqrt{\mu^2 + \mathbf{z}\cdot\mathbf{z}}, \mathbf{z})$ one has $\hat{f}(\mathbf{z}) =$

$\frac{1}{2}(\Psi_{\mathsf{f}}(z_+) + \Psi_{\mathsf{f}}(z_-))$, and similarly for $\hat{g}$, so (38) bounds $\hat{f}$ and $\hat{g}$ on $\mathbb{C}^3$. Since the derivatives of $\hat{f}$ at 0 are the moments of f, Cauchy estimates then bound the moments of a source about a centre $\mathbf{x}_0$,

$$\left|\int (\mathbf{x}-\mathbf{x}_0)^{\alpha} f\right| \le M_b (B_b T)^{|\alpha|} e^{\Lambda_b \mu T} \, \|\mathsf{R}_U^b \mathsf{f}\|,$$

and similarly for g, for every multi-index $\alpha \in \mathbb{N}^3$ and $U \subset B_T(\mathbf{x}_0)$, with constants M_b, B_b, Λ_b depending only on b (Lemma 9, which also keeps a factor $T/(1+\mu T)$ omitted here). Between the two balls the covariance is an integral against a smooth kernel, a Bessel function of $\mathbf{y}-\mathbf{x}$ (Appendix E). For the forward estimate, expand this kernel in a Taylor series in $\mathbf{x}$ and $\mathbf{y}$ about the two centres:

$$\mathsf{v}_\mu(\mathsf{f},\mathsf{f}') = \sum_{\alpha,\beta} c_{\alpha\beta} \int \mathbf{x}^{\alpha} f \int \mathbf{y}^{\beta} f' + (\text{same for } g, g').$$

Each moment of f is bounded by $\|\mathsf{R}_A^b \mathsf{f}\|$ and each moment of f' by $\|\mathsf{R}_B^b \mathsf{f}'\|$, so each term is a rank-one operator of the form $\mathsf{R}_A^b(\cdot)\mathsf{R}_B^b$, and the sum $X_{b,\mu,s}$ of the middle factors converges in trace norm. For the reverse estimate, $\mathsf{D}_{\mu,s}$ has the kernel $-\mathsf{v}_\mu$ between the two balls and 0 within each ball (by (34)), so it is not a single Taylor series. It is approximated by finite-rank operators F_n with polynomial kernels, and each increment $\mathsf{G}_n = \mathsf{F}_{n+1} - \mathsf{F}_n$ satisfies two bounds: a decaying one, with $\|\mathsf{G}_n\|_1$ of order $s^{-2}(c_{\mathrm{pol}}/s)^n$ for a constant c_{pol}, up to factors polynomial in n and bounded in μ, and a weighted one, $\mathsf{G}_n = \mathsf{R}_{U_s}^{1/2} \mathsf{B}_n \mathsf{R}_{U_s}^{1/2}$ with $\|\mathsf{B}_n\|_1$ growing geometrically in n. Interpolating between them (Hadamard's three-lines theorem) gives, for $0 < \theta < 1$,

$$\|\mathsf{R}_{U_s}^{-\theta/2} \mathsf{G}_n \mathsf{R}_{U_s}^{-\theta/2}\|_1 \le \|\mathsf{G}_n\|_1^{1-\theta} \|\mathsf{B}_n\|_1^{\theta}.$$

The growth of $\|\mathsf{B}_n\|_1$ must be overcome by the decay of $\|\mathsf{G}_n\|_1$, which forces θ to be small; the proof takes $\theta = 2^{-17}$, which gives the power $a = \theta/2 = 2^{-18}$ and the decay $s^{-2(1-\theta)} = s^{-\gamma}$.

VI. COMPLETION OF THE PROOF

This section fixes the Rényi orders $r_\pm$; combines Theorems 4 and 5 with Section IV to bound $K_{r_\pm,\pm}$ (49) and the clustering constant, $q_{\mathrm{cl}} = Q_0/s$ (50); checks the cutoff (17) of Theorem 2 for $s \ge C_*$; and so proves Theorem 1 and Corollary 1.

Parameters.—With $a = 2^{-18}$ from Theorem 5, fix

$$\begin{aligned} r_- &= \frac{1}{1-a} = \frac{2^{18}}{2^{18}-1}, \qquad r_+ = \frac{4}{a} = 2^{20}, \\ b &= 1 - \frac{1}{r_+} = 1 - 2^{-20}. \end{aligned} \tag{41}$$

Then $1 < r_- < 2$, r_+ is an integer as (30) requires, $a = 1 - 1/r_-$ and $b = 1 - 1/r_+$ as in (15), and

$$\begin{aligned} \eta &= 1 - \frac{1-b}{ab^2} = 1 - \frac{1}{4b^2} > 0, \\ \nu_+ &= \frac{1+b}{4b^2}, \qquad \nu_- = 2^{18} - 1. \end{aligned} \tag{42}$$

The small reverse exponent forces the large forward order: $\eta > 0$ needs $r_+ > 1/(ab^2)$. With F_b from Theorem 4 and $\mathcal{J}_0$ from Theorem 5, put

$$\varepsilon_+(s) = (r_+ - 1) F_b \, s^{-2}, \qquad \varepsilon_-(s) = 2^{2a-1} \mathcal{J}_0 \, s^{-\gamma}. \tag{43}$$

Forward comparison.—Choose coordinates on $F = F_A \oplus F_B$ in which $V_{\sigma_F} = I$; since $\sigma_F = \rho_{F_A} \otimes \rho_{F_B}$, these are coordinates on F_A and on F_B orthonormal for v_μ, and the inclusions $\iota_A \colon F_A \to \mathcal{X}_{\mu,B_1(0)}$ and $\iota_B \colon F_B \to \mathcal{X}_{\mu,B_1(s\mathbf{e}_1)}$ are isometric. By (33),

$$V_{\rho_F} - V_{\sigma_F} = \begin{pmatrix} 0 & \iota_A^* \mathsf{C}_{\mu,s} \iota_B \\ (\iota_A^* \mathsf{C}_{\mu,s} \iota_B)^* & 0 \end{pmatrix}.$$

The thermal increment of the product σ_F at $\tau = b$ is block diagonal, with blocks $\mathsf{H}_b(I_{F_A})$ and $\mathsf{H}_b(I_{F_B})$, so the matrix Z of Theorem 3 for (ρ_F, σ_F, r_+), which we call $\mathsf{Z}_{+,F}$, has zero diagonal blocks and the off-diagonal block

$$\mathsf{Z}_{AB} = \mathsf{H}_b(I_{F_A})^{-1/2} \, \iota_A^* \mathsf{C}_{\mu,s} \iota_B \, \mathsf{H}_b(I_{F_B})^{-1/2};$$

its norms do not depend on the choice of coordinates (Appendix B). For $s \ge S_b$, (35) and (32) with $r = r_+$ give $\|\mathsf{Z}_{AB}\|_1 \le (r_+ - 1) F_b s^{-2} = \varepsilon_+(s)$, and since the eigenvalues of $\mathsf{Z}_{+,F}$ are $\pm$ the singular values of Z_{AB} and zeros,

$$\|\mathsf{Z}_{+,F}\| \le \varepsilon_+(s), \qquad \|\mathsf{Z}_{+,F}\|_1 \le 2\varepsilon_+(s). \tag{44}$$

Here $k_{r_+} = \max\{2, 1/b\} = 2$, so if $\varepsilon_+(s) \le \frac{1}{8}$, Theorem 3 gives

$$\log \widetilde{Q}_{r_+}(\rho_F \| \sigma_F) \le 8 r_+ \varepsilon_+(s). \tag{45}$$

Reverse comparison.—Now choose coordinates on F in which $V_{\rho_F} = I$, that is, orthonormal for the joint covariance v_μ, so that the inclusion $\iota_F \colon F \to \mathcal{X}_{\mu,U_s}$ is isometric. By (34), $V_{\sigma_F} - V_{\rho_F} = \iota_F^* \mathsf{D}_{\mu,s} \iota_F$, so the matrix Z of Theorem 3 for (σ_F, ρ_F, r_-) is

$$\mathsf{Z}_{-,F} = \mathsf{H}_a(I_F)^{-1/2} \, \iota_F^* \mathsf{D}_{\mu,s} \iota_F \, \mathsf{H}_a(I_F)^{-1/2},$$

and for $s \ge S_{\mathrm{rev}}$, (36) and (31) with $\mathsf{R} = \mathsf{R}_{U_s}$ give

$$\|\mathsf{Z}_{-,F}\|_1 \le 2^{2a-1} \mathcal{J}_0 s^{-\gamma} = \varepsilon_-(s). \tag{46}$$

Here $k_{r_-} = 1/a$, so if $\varepsilon_-(s) \le a/4$,

$$\log \widetilde{Q}_{r_-}(\sigma_F \| \rho_F) \le \frac{2r_-}{a} \varepsilon_-(s). \tag{47}$$

Neither bound depends on F_A, F_B or the temperatures β_j of their modes.

All local events.—Inserting (45) and (47) into (22),

$$\begin{aligned} \omega(EP)^{r_+} &\le e^{8 r_+ \varepsilon_+(s)} \big(\omega(E)\omega(P)\big)^{r_+ - 1}, \\ \big(\omega(E)\omega(P)\big)^{r_-} &\le e^{2 r_- \varepsilon_-(s)/a} \, \omega(EP)^{r_- - 1}, \end{aligned} \tag{48}$$

for effects of the finite algebras. Both are free of denominators and continuous in the three probabilities, with constants

depending only on s, so Lemma 3 extends them to all $E \in \mathfrak{A}$, $P \in \mathfrak{B}$; for projections with $\omega(E)\omega(P) > 0$ they bound the quotients in (13), the second forcing $\omega(EP) > 0$. Hence the event comparison constants satisfy

$$\log K_{r_+,+} \le 8r_+\varepsilon_+(s), \qquad \log K_{r_-,-} \le \frac{2r_-}{a}\varepsilon_-(s). \tag{49}$$

For the correlations, (21) with $r_+ \ge 2$ and (45) give

$$\|\rho_F - \sigma_F\|_1 \le \sqrt{\exp\Big(\frac{8\varepsilon_+(s)}{b}\Big) - 1},$$

using $r_+/(r_+ - 1) = 1/b$; for contractions A, B of the finite algebras, $\omega(AB) - \omega(A)\omega(B) = \mathrm{Tr}((\rho_F - \sigma_F)AB)$ is bounded by this, and Lemma 3 extends the bound to $\mathfrak{A}, \mathfrak{B}$. If $\varepsilon_+(s) \le b/8$, then $e^x - 1 \le 2x$ on $[0, 1]$ gives

$$|\omega(AB) - \omega(A)\omega(B)| \le \frac{Q_0}{s}, \qquad Q_0 = 4\sqrt{(r_+ - 1)F_b/b}. \tag{50}$$

The constant.—Define

$$C_{\mathrm{prob}} = Q_0 \exp\Big(\frac{2\log 2/(ab) + \nu_+ + \nu_-}{\eta}\Big), \tag{51}$$

$$C_* = \max\Big\{5,\ S_b,\ S_{\mathrm{rev}},\ \sqrt{8r_+(r_+ - 1)F_b},\ \Big(\frac{2^{2a+1}\mathcal{J}_0}{a}\Big)^{1/\gamma},\ C_{\mathrm{prob}}\Big\}. \tag{52}$$

Every entry is finite and independent of μ. For the parameters (41), $C_{\mathrm{prob}} = Q_0\, 10^{362\,232.284166\ldots}$; the power of ten is an explicit multiplier relative to Q_0, not a numerical determination of C_*: F_b, $\mathcal{J}_0$ and hence Q_0 are defined through finite covers and approximation bounds and are known only to be finite. The first entry gives $C_* \ge 5 > 2$ independently of Q_0, and the proof of Theorem 4 takes $S_b \ge (4 + 80\sqrt{3})\,\Lambda_b$ with $\Lambda_b = 2^{15}/\sin(\pi b)$ as in (D3) (Appendix E), which at $b = 1 - 2^{-20}$ exceeds 1.5×10^{12}.

Proof of Theorem 1. Fix $\mu > 0$ and $s \ge C_*$. Then $s \ge 5 > 2$, $s \ge S_b$ and $s \ge S_{\mathrm{rev}}$, so (44) and (46) hold for all finite systems. From $s^2 \ge 8r_+(r_+ - 1)F_b$, $\varepsilon_+(s) \le 1/(8r_+) \le b/8 < \frac{1}{8}$, using $r_+ b = r_+ - 1 \ge 1$; from $s^\gamma \ge 2^{2a+1}\mathcal{J}_0/a$, $\varepsilon_-(s) \le a/4$. Hence (45), (47), (49) and (50) hold, with $\log K_{r_+,+} \le 1$ and $\log K_{r_-,-} \le r_-/2 < 1$. Take $q_{\mathrm{cl}} = Q_0/s$; since $s \ge C_{\mathrm{prob}}$,

$$\Big(\frac{Q_0}{s}\Big)^\eta K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} \le \Big(\frac{Q_0}{C_{\mathrm{prob}}}\Big)^\eta e^{\nu_+ + \nu_-} = 2^{-2/(ab)},$$

which is (17). The algebras $\mathfrak{A}, \mathfrak{B}$ commute, $r_+ > 2$, $1 < r_- < 2$ and $\eta > 0$, so Theorem 2 gives $\beta_\mu(1, s) = 1$. For $m, R > 0$ and $L \ge C_* R$, Lemma 1 gives $\beta_m(R, L) = \beta_{mR}(1, L/R) = 1$, and C_* does not depend on m, R. □

Proof of Corollary 1. Let e be the unit vector of the given direction and $\mathcal{L}$ a Lorentz transformation with $\mathcal{L}e = (0, \mathbf{e}_1)$. The vacuum is Poincaré invariant, and the unitary implementing $\mathcal{L}$ carries $\mathfrak{A}_m(O)$ onto $\mathfrak{A}_m(\mathcal{L}O)$, so the Bell correlation of $\mathfrak{A}_m(O_1)$ and $\mathfrak{A}_m(O_2 + Le)$ equals that of $\mathfrak{A}_m(\mathcal{L}O_1)$ and $\mathfrak{A}_m(\mathcal{L}O_2 + (0, L\mathbf{e}_1))$. Choose R with $\mathcal{L}O_1 \cup \mathcal{L}O_2 \subset D_R$. By isotony, $O \subset O'$ implies $\mathfrak{A}_m(O) \subset \mathfrak{A}_m(O')$, the two algebras lie in $\mathfrak{A}_m(D_R)$ and $\mathfrak{A}_m(D_R + (0, L\mathbf{e}_1))$, which commute for $L > 2R$, and enlarging both algebras can only increase the supremum (1); so the Bell correlation is at most $\beta_m(R, L) = 1$ for $L \ge C_* R$, and it is always at least 1. □

Appendix A: Proof of the probability criterion

This appendix proves Theorem 2.

Step 1: reduction to reflections. The choice $A_i = B_j = I$ gives the value 1 in (1), so $\beta(\omega; \mathfrak{A}, \mathfrak{B}) = 1$ unless some self-adjoint contractions $A_0, A_1 \in \mathfrak{A}$ and $B_0, B_1 \in \mathfrak{B}$ give the CHSH expression an absolute value larger than 1. The expression is real, because the A_i and B_j commute, and negating both observables in one of the two regions changes its sign, so we may assume a value larger than 1. Every self-adjoint contraction is a norm-convergent average of reflections, $A = \frac{1}{2}\int_{-1}^{1}(2\mathbf{1}_{\{A \ge u\}} - I)\,\mathrm{d}u$, where $\mathbf{1}_{\{A \ge u\}}$ is the spectral projection of A for the interval $[u, 1]$, and these reflections lie in every von Neumann algebra that contains A. The CHSH expression is affine and norm-continuous in each of its four arguments, so a value larger than 1 at A_0 gives a value larger than 1 at one of these reflections; replacing A_0, A_1, B_0, B_1 in turn, a strict violation by contractions becomes a strict violation by four reflections. For reflections, the *score* is $S = \frac{1}{2}\omega\big(A_0(B_0 + B_1) + A_1(B_0 - B_1)\big)$.

Step 2: cells and their symmetries. With $E_i^\pm = (I \pm A_i)/2$ and $P_j^\pm = (I \pm B_j)/2$, which are projections, the sixteen *cells* are the joint probabilities $\omega(E_i^\pm P_j^\pm)$; the index pair (i, j) is called the *setting pair*. A cell of the setting pair (i, j) is *winning* when the product of its two outcomes equals the sign with which $\omega(A_i B_j)$ enters $S = \frac{1}{2}(\omega(A_0B_0) + \omega(A_0B_1) + \omega(A_1B_0) - \omega(A_1B_1))$; so the winning cells are $E_i^+P_j^+$ and $E_i^-P_j^-$ for $(i, j) \ne (1, 1)$, and $E_1^+P_1^-$ and $E_1^-P_1^+$ for $(i, j) = (1, 1)$. Since $\pm\omega(A_iB_j)$, with the sign it carries in S, equals the sum of the two winning cells of (i, j) minus the sum of the two losing ones, and the four cells of each setting pair add up to 1, the value $S + 2$ is the sum of the eight winning cells. The three replacements $(A_0, A_1, B_0, B_1) \mapsto (A_0, -A_1, B_1, B_0)$, $(A_1, A_0, B_0, -B_1)$ and $(-A_0, -A_1, -B_0, -B_1)$ carry reflection quartets to reflection quartets, leave S unchanged, and permute the eight winning cells, and every winning cell can be moved to every other by composing them: the first two exchange the settings at one site and flip one outcome to compensate the sign of the $(1, 1)$ correlator, and the third exchanges the two winning cells of each setting pair. We may therefore assume that a least likely winning cell is $w = \omega(E_0^-P_0^-)$.

Step 3: the probability form of $S > 1$. Write $e_A = \omega(E_0^-)$, $e_B = \omega(P_0^-)$, $x_A = \omega(E_1^-)$, $x_B = \omega(P_1^-)$. Since $A_i = I - 2E_i^-$ and $B_j = I - 2P_j^-$, and $\omega((I - 2E)(I - 2P)) = 1 - 2\omega(E) - 2\omega(P) + 4\omega(EP)$, the four correlators are $1 - 2e_A - 2e_B + 4w$, $1 - 2e_A - 2x_B + 4\omega(E_0^-P_1^-)$, $1 - 2x_A - 2e_B + 4\omega(E_1^-P_0^-)$ and $1 - 2x_A - 2x_B + 4\omega(E_1^-P_1^-)$. Inserting them into S, the terms in x_A, x_B cancel, and using $\omega(E_0^-P_1^+) = e_A - \omega(E_0^-P_1^-)$ and

$\omega(E_1^+P_0^-) = e_B - \omega(E_1^-P_0^-)$,

$$S = 1 + 2\big\{w - \omega(E_0^-P_1^+) - \omega(E_1^+P_0^-) - \omega(E_1^-P_1^-)\big\}.$$

So $S > 1$ reads

$$w > \omega(E_0^-P_1^+) + \omega(E_1^+P_0^-) + \omega(E_1^-P_1^-). \tag{A1}$$

The three terms on the right are nonnegative, so $w > 0$ and each of them is smaller than w.

Step 4: the neighbouring winning cells. The cells $E_0^-P_1^-$ and $E_1^-P_0^-$ are winning, so their probabilities are at least w; by Step 3 they are also $e_A - \omega(E_0^-P_1^+) > e_A - w$ and $e_B - \omega(E_1^+P_0^-) > e_B - w$. Hence $\omega(E_0^-P_1^-) \geq \max\{w, e_A - w\} \geq e_A/2$ and $\omega(E_1^-P_0^-) \geq e_B/2$.

Step 5: the marginals x_A, x_B. With the exponents a, b of (15), abbreviate $\overline{k} = K_{r_+,+}^{-1/r_+}$ and $\underline{k} = K_{r_-,-}^{-1/(r_--1)}$, so that (14) reads $\omega(EP) \leq \overline{k}(\omega(E)\omega(P))^b$ and $\omega(EP) \geq \underline{k}(\omega(E)\omega(P))^{1/a}$. The upper bound at the two neighbouring cells gives $e_A/2 \leq \overline{k}(e_Ax_B)^b$ and $e_B/2 \leq \overline{k}(x_Ae_B)^b$. Since $e_A \geq w > 0$ and $e_B \geq w > 0$, dividing and taking the power $1/b$,

$$x_B \geq (2\overline{k})^{-1/b} e_A^{(1-b)/b}, \qquad x_A \geq (2\overline{k})^{-1/b} e_B^{(1-b)/b}.$$

Step 6: a lower bound on w. The upper bound at $E_0^-P_0^-$ reads $w \leq \overline{k}(e_Ae_B)^b$, that is, $e_Ae_B \geq (w/\overline{k})^{1/b}$. By (A1), the lower bound in (14) at $E_1^-P_1^-$, Step 5, and this inequality raised to the positive power $(1-b)/(ab)$,

$$\begin{aligned} w > \omega(E_1^-P_1^-) &\geq \underline{k}(x_Ax_B)^{1/a} \\ &\geq \underline{k}(2\overline{k})^{-2/(ab)}(e_Ae_B)^{(1-b)/(ab)} \\ &\geq \underline{k}\,2^{-2/(ab)}\overline{k}^{-(1+b)/(ab^2)} w^{(1-b)/(ab^2)}, \end{aligned}$$

the exponents of $\overline{k}$ adding up to $-2/(ab) - (1-b)/(ab^2) = -(1+b)/(ab^2)$. Dividing by $w^{(1-b)/(ab^2)}$ and using

$$\overline{k}^{-(1+b)/(ab^2)} = K_{r_+,+}^{-\nu_+} \quad \text{and} \quad \underline{k} = K_{r_-,-}^{-1/(r_--1)} = K_{r_-,-}^{-\nu_-},$$

which hold by (15) because $r_- - 1 = a/(1-a)$,

$$w^\eta K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} > 2^{-2/(ab)}. \tag{A2}$$

Step 7: an upper bound on w from clustering. Let S_{prod} be S with each $\omega(A_iB_j)$ replaced by $\omega(A_i)\omega(B_j)$. Since $|\omega(A_i)| \leq 1$ and $\frac{1}{2}(|x+y|+|x-y|) = \max\{|x|,|y|\}$, $S_{\mathrm{prod}} \leq \max_j|\omega(B_j)|$. The bound (16) on the four pairs gives $|S - S_{\mathrm{prod}}| \leq 2q_{\mathrm{cl}}$, so $1 < S \leq \max_j|\omega(B_j)| + 2q_{\mathrm{cl}}$. For the j attaining the maximum, the less probable outcome of B_j has probability $(1 - |\omega(B_j)|)/2 < q_{\mathrm{cl}}$. If that outcome is $P_j^\pm$, the cell $E_0^\pm P_j^\pm$ is winning, because $\omega(A_0B_j)$ enters S with the sign $+$, and $E_0^\pm P_j^\pm \leq P_j^\pm$; so $w \leq \omega(P_j^\pm) < q_{\mathrm{cl}}$.

Conclusion. Since $\eta > 0$, $0 < w < q_{\mathrm{cl}}$ and $K_{r_\pm,\pm} \geq 1$, we get $w^\eta K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} < q_{\mathrm{cl}}^\eta K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} \leq 2^{-2/(ab)}$ by (17), contradicting (A2). Hence no reflection quartet has $S > 1$, and $\beta(\omega; \mathfrak{A}, \mathfrak{B}) = 1$ by Step 1. □

Appendix B: Proofs of the Gaussian comparison and of its consequences

Gaussian conventions.—For n modes with commutation matrix J_n, a centred Gaussian state ρ_V with covariance V has characteristic function $\mathrm{Tr}(\rho_V e^{ix^T(q,p)}) = e^{-x^TVx/2}$, $x \in \mathbb{R}^{2n}$, where $(q,p) = (\Phi(\mathsf{f}_1), \ldots, \Phi(\mathsf{f}_{2n}))$ is the column of canonical operators in a basis as in Section II, in the normalization where the oscillator ground state has covariance $I/2$ [17]. For faithful σ with covariance V_σ, abbreviate $\mathsf{T}_\tau = V_\sigma^{[\tau]}$ and $\mathsf{L} = (\mathsf{T}_\tau^2 - I/4)^{1/2}$, and work in Williamson normal-mode coordinates of σ, $V_\sigma = \bigoplus_j \upsilon_j I_2$: the trace formula below is coordinate independent, but the covariance formula holds only there. For $\Theta = \sigma^{-\tau/2}\rho\,\sigma^{-\tau/2}$, the two Gaussian sandwich formulas

$$\mathrm{Tr}\,\Theta = \frac{\mathrm{Tr}\,\sigma^{1-\tau}}{\sqrt{\det(I-\mathsf{Z})}}, \qquad V_\Theta = -\mathsf{T}_\tau + \mathsf{L}(\mathsf{T}_\tau - V_\rho)^{-1}\mathsf{L}, \tag{B1}$$

hold whenever $V_\rho < \mathsf{T}_\tau$ (verified in Step 2 below), where V_Θ is the covariance of the Gaussian state $\Theta/\mathrm{Tr}\,\Theta$ and Z is the matrix of Theorem 3. The trace formula is that of Seshadreesan, Lami and Wilde [17, Theorem 22], who compute $\widetilde{Q}_r$ for Gaussian states from the opposite sandwich $\rho^{1/2}\sigma^{-\tau}\rho^{1/2}$; it transfers because $\rho^{1/2}\sigma^{-\tau}\rho^{1/2} = (\sigma^{-\tau/2}\rho^{1/2})^*(\sigma^{-\tau/2}\rho^{1/2})$ and $\Theta = (\sigma^{-\tau/2}\rho^{1/2})(\sigma^{-\tau/2}\rho^{1/2})^*$ have the same nonzero spectrum. The covariance formula is specific to the sandwich used here; both follow from the coherent-state (Bargmann) representation [32]. By (11), the jth mode of σ is $(1-e^{-\beta_j})e^{-\beta_jN_j}$, N_j its number operator, and t^{N_j} acts on the Bargmann space of entire functions by scaling the jth variable with t; so Θ has the integral kernel of ρ with both arguments scaled by $\mathbb{T} = \bigoplus_j e^{\tau\beta_j/2}I_2$, times $\prod_j(1-e^{-\beta_j})^{-\tau}$. The kernel of ρ is the exponential of a quadratic form determined by V_ρ, and the scaling keeps it Gaussian, so the trace and the characteristic function of Θ are Gaussian integrals; they converge, and Θ is trace class, exactly when $V_\rho < \mathsf{T}_\tau$, and a Schur complement yields (B1) through the mode identities

$$\mathsf{T}_\tau - \tfrac{1}{2}I = (\mathbb{T}^2 - I)^{-1}, \qquad \mathsf{L} = \mathbb{T}\,(\mathbb{T}^2 - I)^{-1}.$$

Proof of Theorem 3. Step 1: coordinate independence. For a real invertible S, a positive Γ and a real symmetric Ξ, the matrix $(S\Gamma S^T)^{-1/2}S\Gamma^{1/2}$ is orthogonal, so $(S\Gamma S^T)^{-1/2}\,S\Xi S^T\,(S\Gamma S^T)^{-1/2}$ is orthogonally equivalent to $\Gamma^{-1/2}\Xi\Gamma^{-1/2}$. A real linear change of the canonical coordinates replaces V_ρ, V_σ and $\mathsf{H}_\tau(V_\sigma)$ by $SV_\rho S^T$ and so on, so with $\Gamma = \mathsf{H}_\tau(V_\sigma)$ and $\Xi = V_\rho - V_\sigma$ it replaces Z by an orthogonally equivalent matrix. Hence $\|\mathsf{Z}\|$, $\|\mathsf{Z}\|_1$ and the hypothesis are coordinate independent, as is $\widetilde{Q}_r(\rho\|\sigma)$. We work in Williamson coordinates of σ, in which V_σ, T_τ, L and $\mathsf{T}_{1-\tau} = V_\sigma^{[1-\tau]}$ are all diagonal.

Step 2: the condition $V_\rho < \mathsf{T}_\tau$. With $\delta = \|\mathsf{Z}\| \leq (4k_r)^{-1} \leq \frac{1}{8}$, $\mathsf{T}_\tau - V_\rho = \mathsf{H}_\tau(V_\sigma) - (V_\rho - V_\sigma) = \mathsf{H}_\tau(V_\sigma)^{1/2}(I - \mathsf{Z})\mathsf{H}_\tau(V_\sigma)^{1/2} > 0$, so (B1) applies.

Step 3: the perturbed covariance. At $V_\rho = V_\sigma$ the sandwich is $\sigma^{1-\tau}$, whose normalized state has covariance $\mathsf{T}_{1-\tau}$, so (B1)

gives $\mathsf{T}_{1-\tau} = -\mathsf{T}_\tau + \mathsf{L}\,\mathsf{H}_\tau(V_\sigma)^{-1}\mathsf{L}$. Subtracting this from (B1) and using Step 2,

$$V_\Theta - \mathsf{T}_{1-\tau} = \mathsf{P}\big((I-\mathsf{Z})^{-1} - I\big)\mathsf{P},$$
$$\mathsf{P} = \mathsf{L}\,\mathsf{H}_\tau(V_\sigma)^{-1/2}, \qquad \mathsf{P}^2 = \mathsf{T}_{1-\tau} + \mathsf{T}_\tau.$$

Mode by mode, with $u = \operatorname{artanh}(1/(2\upsilon_j))$, the entries of T_τ and $\mathsf{T}_{1-\tau}$ are $\frac{1}{2}\coth(\tau u)$ and $\frac{1}{2}\coth((1-\tau)u)$ by (23); since $x\coth x$ increases and coth decreases, $\coth(\tau u)/\coth((1-\tau)u) \le \max\{1,(1-\tau)/\tau\}$, so $\mathsf{P}^2 \le k_r\mathsf{T}_{1-\tau}$. The relative change of the covariance of Θ, the analogue of Z for Θ, is

$$\begin{aligned}\mathsf{Z}_\Theta &= \mathsf{T}_{1-\tau}^{-1/2}(V_\Theta - \mathsf{T}_{1-\tau})\mathsf{T}_{1-\tau}^{-1/2}\\ &= (\mathsf{T}_{1-\tau}^{-1/2}\mathsf{P})\,\mathsf{Z}(I-\mathsf{Z})^{-1}\,(\mathsf{P}\mathsf{T}_{1-\tau}^{-1/2}).\end{aligned}$$

Since the eigenvalues of Z lie in $[-\delta,\delta]$ and $\|\mathsf{T}_{1-\tau}^{-1/2}\mathsf{P}\|^2 \le k_r$,

$$\|\mathsf{Z}_\Theta\| \le \frac{k_r\delta}{1-\delta} \le \frac{2}{7}, \qquad \|\mathsf{Z}_\Theta\|_1 \le \frac{k_r}{1-\delta}\|\mathsf{Z}\|_1. \tag{B2}$$

Step 4: the purity function. For a *physical* covariance V, one with $V + \frac{\mathrm{i}}{2}J_n \ge 0$, with symplectic eigenvalues υ_j, let $\mathscr{P}_r(V) = \log\operatorname{Tr}\rho_V^r = -\sum_j \log f_r(\upsilon_j)$ with $f_r(x) = (x+\frac{1}{2})^r - (x-\frac{1}{2})^r$; the second expression holds mode by mode, since a thermal mode is $(1-e^{-\beta})e^{-\beta N}$, N the number operator, with $e^{-\beta} = (\upsilon-\frac{1}{2})/(\upsilon+\frac{1}{2})$ by (11), and $\operatorname{Tr}((1-e^{-\beta})e^{-\beta N})^r = 1/f_r(\upsilon)$. One checks $0 \le x f_r'(x)/f_r(x) \le r$ for $x \ge \frac{1}{2}$. The symplectic eigenvalues are the positive eigenvalues of $\mathrm{i}J_nV$; for a function F of the covariance write $\mathrm{d}F(V)[\dot V] = \frac{\mathrm{d}}{\mathrm{d}t}F(V+t\dot V)\big|_{t=0}$, $\dot V$ symmetric. First-order perturbation at $\Upsilon = \bigoplus_j \upsilon_j I_2$ with distinct υ_j gives $\mathrm{d}\upsilon_j[\dot V] = \frac{1}{2}\operatorname{Tr}\dot V_{jj}$ for the jth 2×2 diagonal block, whence $|\,\mathrm{d}\mathscr{P}_r(\Upsilon)[\dot V]| \le \frac{r}{2}\sum_j|\operatorname{Tr}(\upsilon_j^{-1}\dot V_{jj})| \le \frac{r}{2}\|\Upsilon^{-1/2}\dot V\Upsilon^{-1/2}\|_1$. By the invariance of $\mathscr{P}_r$ under $V \mapsto SVS^T$ with S symplectic, and by Step 1, at every faithful covariance

$$|\,\mathrm{d}\mathscr{P}_r(V)[\dot V]| \le \frac{r}{2}\|V^{-1/2}\dot V V^{-1/2}\|_1, \tag{B3}$$

first where the υ_j are distinct, then everywhere by continuity, since $\mathscr{P}_r(V) = -\frac{1}{2}\operatorname{Tr}\log f_r(|V^{1/2}\mathrm{i}J_nV^{1/2}|)$ is smooth on faithful covariances.

Step 5: integration. The segment $V(v) = \mathsf{T}_{1-\tau}^{1/2}(I + v\mathsf{Z}_\Theta)\mathsf{T}_{1-\tau}^{1/2}$, $0 \le v \le 1$, joins $\mathsf{T}_{1-\tau}$ to V_Θ through physical covariances, faithful for $v<1$ because $\mathsf{T}_{1-\tau} + \frac{\mathrm{i}}{2}J_n > 0$. Its derivative is $\mathsf{T}_{1-\tau}^{1/2}\mathsf{Z}_\Theta\mathsf{T}_{1-\tau}^{1/2}$, and by Step 1 the relative perturbation $V(v)^{-1/2}\dot V V(v)^{-1/2}$ has the trace norm of $\mathsf{Z}_\Theta(I+v\mathsf{Z}_\Theta)^{-1}$, at most $\|\mathsf{Z}_\Theta\|_1/(1-\|\mathsf{Z}_\Theta\|)$. Integrating (B3) and letting $v\to 1$,

$$|\mathscr{P}_r(V_\Theta) - \mathscr{P}_r(\mathsf{T}_{1-\tau})| \le \frac{r}{2}\frac{\|\mathsf{Z}_\Theta\|_1}{1-\|\mathsf{Z}_\Theta\|} \le \frac{r}{2}\frac{k_r\|\mathsf{Z}\|_1}{(1-\delta)(1-\|\mathsf{Z}_\Theta\|)}.$$

Step 6: assembly. Since $\Theta/\operatorname{Tr}\Theta$ is Gaussian with covariance V_Θ, (19) and (B1) give $\widetilde{Q}_r(\rho\|\sigma) = (\operatorname{Tr}\Theta)^r e^{\mathscr{P}_r(V_\Theta)} = (\operatorname{Tr}\sigma^{1-\tau})^r\det(I-\mathsf{Z})^{-r/2}e^{\mathscr{P}_r(V_\Theta)}$, and $(\operatorname{Tr}\sigma^{1/r})^r e^{\mathscr{P}_r(\mathsf{T}_{1-\tau})} = \operatorname{Tr}\sigma = 1$ because $\sigma^{1/r}/\operatorname{Tr}\sigma^{1/r}$ has covariance $\mathsf{T}_{1-\tau}$ and $r(1-\tau)=1$. Hence

$$\log\widetilde{Q}_r(\rho\|\sigma) = -\frac{r}{2}\operatorname{Tr}\log(I-\mathsf{Z}) + \mathscr{P}_r(V_\Theta) - \mathscr{P}_r(\mathsf{T}_{1-\tau}).$$

The bound $|\log(1-t)| \le |t|/(1-|t|)$ on the eigenvalues of Z gives $|\operatorname{Tr}\log(I-\mathsf{Z})| \le \|\mathsf{Z}\|_1/(1-\delta)$. With Step 5, $1/(1-\delta) \le 8/7$ and $1/(1-\|\mathsf{Z}_\Theta\|) \le 7/5$,

$$\log\widetilde{Q}_r(\rho\|\sigma) \le r\Big(\frac{4}{7} + \frac{4}{5}k_r\Big)\|\mathsf{Z}\|_1 \le 2rk_r\|\mathsf{Z}\|_1,$$

the last step because $k_r \ge 2$. □

Proof of Lemma 4. Both parts are known: (20) is the two-outcome case of the data-processing inequality for the sandwiched divergence, proved for all $r>1$ by Frank and Lieb [25] and by Beigi [26] and used in this form in strong-converse arguments [23], and at $r=2$ the quantity $\widetilde{Q}_2(\rho\|\sigma)-1$ is a quantum χ^2-divergence, which dominates the squared trace distance [27]. We include the short proofs.

The effect bound. Finiteness of $\widetilde{Q}_r(\rho\|\sigma)$ means that $\Theta = \sigma^{-\tau/2}\rho\sigma^{-\tau/2}$ has finite r-norm $\|\Theta\|_r = \widetilde{Q}_r(\rho\|\sigma)^{1/r}$, and $\rho = \sigma^{\tau/2}\Theta\sigma^{\tau/2}$. Since $\tau + 1/r = 1$, Hölder's inequality for Schatten norms, $|\operatorname{Tr}(XY)| \le \|X\|_p\|Y\|_q$ for $1/p+1/q=1$ [33], gives $\rho(M) = \operatorname{Tr}(\Theta\,\sigma^{\tau/2}M\sigma^{\tau/2}) \le \|\Theta\|_r\|\sigma^{\tau/2}M\sigma^{\tau/2}\|_{1/\tau}$. The operator $\sigma^{\tau/2}M\sigma^{\tau/2}$ has the same nonzero spectrum as $M^{1/2}\sigma^\tau M^{1/2}$, and Hansen's inequality for the operator monotone function $t\mapsto t^\tau$ (monotone: $0\le X\le Y$ implies $X^\tau \le Y^\tau$) and the contraction $M^{1/2}$ [34] gives $M^{1/2}\sigma^\tau M^{1/2} \le (M^{1/2}\sigma M^{1/2})^\tau$; so the second factor is at most $(\operatorname{Tr}M^{1/2}\sigma M^{1/2})^\tau = \sigma(M)^\tau$. Raising $\rho(M) \le \widetilde{Q}_r^{1/r}\sigma(M)^\tau$ to the power r gives (20).

Comparison of the orders 2 *and* r. For $r\ge 2$, interpolation by Hadamard's three-lines theorem for Schatten norms [33], applied to $z\mapsto \sigma^{-\tau z/2}\rho\sigma^{-\tau z/2}$ on the strip $0\le\operatorname{Re}z\le 1$, between the trace norm at $\operatorname{Re}z=0$ (where the sandwiching factors are unitary) and the r-norm at $\operatorname{Re}z=1$, evaluated at $z=1/(2\tau)\in(0,1]$ where the interpolated exponent is 2, gives $\widetilde{Q}_2(\rho\|\sigma) \le \widetilde{Q}_r(\rho\|\sigma)^{1/(r-1)}$. To justify the interpolation for the unbounded inverse powers, restrict ρ,σ to the spectral subspace of σ for $[\epsilon,1]$, where σ^{-1} is bounded, and let $\epsilon\downarrow 0$. This is the monotonicity of the sandwiched divergence in its order [26, Theorem 7], [24, Theorem 7].

The trace distance. With $\Theta_{1/2} = \sigma^{-1/4}\rho\sigma^{-1/4}$, the sandwich Θ at $r=2$, i.e. $\tau=\frac{1}{2}$, one has $\operatorname{Tr}(\Theta_{1/2}\sigma^{1/2}) = \operatorname{Tr}\rho = 1$, so $\|\Theta_{1/2}-\sigma^{1/2}\|_2^2 = \widetilde{Q}_2(\rho\|\sigma)-1$, and $\rho-\sigma = \sigma^{1/4}(\Theta_{1/2}-\sigma^{1/2})\sigma^{1/4}$ with $\|\sigma^{1/4}\|_4 = 1$. Hölder's inequality gives $\|\rho-\sigma\|_1 \le \sqrt{\widetilde{Q}_2(\rho\|\sigma)-1}$, and the order comparison gives (21). □

Appendix C: Thermal weights and compression

Proof of Lemma 5. By the functional calculus of $d = |\frac{\mathrm{i}}{2}\mathsf{J}|$, as in (25), it suffices to show that $\mathfrak{q}_\tau(d) = (d\coth(\tau\operatorname{artanh}d) - 1)(1-d^2)^{-\tau} \ge c_\tau$ for $0<d<1$; at $d=0$ and $d=1$, which the spectrum may reach on the full source space, (26) means the continuous extension (values $(1-\tau)/\tau$ and 0), and the bound follows by continuity. Write $d=\tanh u$, $u>0$ (on mode j of a finite system, $u=\beta_j/2$). Then $(1-d^2)^{-\tau} = \cosh(u)^{2\tau}$

and $\tanh u\coth(\tau u)-1=\sinh((1-\tau)u)/(\cosh u\sinh(\tau u))$, so

$$\mathfrak{q}_\tau(\tanh u)=\frac{\sinh((1-\tau)u)}{\sinh(\tau u)}\cosh(u)^{2\tau-1},$$

with limits $(1-\tau)/\tau$ at $u\to 0$ and $2^{1-2\tau}$ at $u\to\infty$. For fixed u the function $\mathfrak{h}(v)=v\coth(vu)$ is convex in v, with $\mathfrak{h}'(v)\le\mathfrak{h}'(1)\le\tanh u$ for $0<v\le 1$, the last inequality being $\tanh u\le u$. By the mean value theorem, $\frac{\mathrm{d}}{\mathrm{d}u}\log\mathfrak{q}_\tau(\tanh u)=\mathfrak{h}(1-\tau)-\mathfrak{h}(\tau)+(2\tau-1)\tanh u=(1-2\tau)(\mathfrak{h}'(v_0)-\tanh u)$ at some v_0 between τ and $1-\tau$, which is ≤ 0 for $\tau\le\frac12$ and ≥ 0 for $\tau\ge\frac12$. So $\mathfrak{q}_\tau(\tanh u)$ is monotone in u, and its infimum is the limit at infinity for $\tau\le\frac12$ and the limit at zero for $\tau\ge\frac12$, which is c_τ. □

Proof of Lemma 6. Step 1: compression of parallel sums and geometric means. For positive invertible A,B, the *parallel sum* $(A^{-1}+B^{-1})^{-1}$ has the variational formula $\langle z,(A^{-1}+B^{-1})^{-1}z\rangle=\inf_{x+y=z}\{\langle x,Ax\rangle+\langle y,By\rangle\}$, which shows that restricting x,y to the range of ι increases the infimum, so $\iota^*(A^{-1}+B^{-1})^{-1}\iota\le((\iota^*A\iota)^{-1}+(\iota^*B\iota)^{-1})^{-1}$, and likewise for more than two summands. The weighted geometric mean $A\#_\tau B=A^{1/2}(A^{-1/2}BA^{-1/2})^\tau A^{1/2}$ has the integral representation $\frac{\sin\pi\tau}{\pi}\int_0^\infty v^{\tau-1}(A^{-1}+vB^{-1})^{-1}\,\mathrm{d}v$ through parallel sums, so $\iota^*(A\#_\tau B)\iota\le(\iota^*A\iota)\#_\tau(\iota^*B\iota)$ [35].

Step 2: the weight operator. G and $\overline{G}$ commute, before and after compression, because their sum is $2I$ and $\iota^*(2I)\iota=2I_F$. For commuting operators $A\#_{1/2}B=(AB)^{1/2}$, so Step 1 at $\tau=\frac12$ gives $\mathsf{R}_F\ge\iota^*\mathsf{R}\iota$. Since $x\mapsto x^{2a}$ is operator monotone ($0\le X\le Y$ implies $X^{2a}\le Y^{2a}$) and operator concave on $[0,\infty)$ for $0<2a\le 1$, $\mathsf{R}_F^{2a}\ge(\iota^*\mathsf{R}\iota)^{2a}\ge\iota^*\mathsf{R}^{2a}\iota$, the second step being Jensen's inequality $\iota^*X^{2a}\iota\le(\iota^*X\iota)^{2a}$ for the isometry ι [34]. This is (29).

Step 3: the thermal weight. For an integer $r\ge 2$, the thermal increment is a parallel sum of geometric means,

$$\mathsf{H}_{(r-1)/r}(I)=\Big[\sum_{k=1}^{r-1}(G\#_{k/r}\overline{G})^{-1}\Big]^{-1}\tag{C1}$$

(for noninvertible $G,\overline{G}$ both sides are decreasing limits over $G+\epsilon I,\overline{G}+\epsilon I$; Step 4). Indeed, at a spectral value $d=\tanh u$ of $\frac{\mathrm{i}}{2}\mathsf{J}$—here d is signed, but both sides of (C1) are even in it, so $u>0$ suffices—one has $G=1+d$, $\overline{G}=1-d$ and $G\#_{k/r}\overline{G}=(1+d)e^{-2uk/r}$, so the right-hand side equals

$$\frac{(1+d)e^{-2u(r-1)/r}\big(1-e^{-2u/r}\big)}{1-e^{-2u(r-1)/r}}=\frac{\sinh(u/r)}{\cosh u\,\sinh((r-1)u/r)},$$

which is $d\coth(\frac{r-1}{r}\operatorname{artanh}d)-1$ by the computation in the previous proof. Compressing each geometric mean and then the parallel sum by Step 1, and using (C1) on F, whose own $G,\overline{G}$ are $G_F,\overline{G}_F$, gives $\mathsf{H}_b(I_F)\ge\iota^*\mathsf{H}_b(I)\iota$, and Lemma 5 at $\tau=(r-1)/r\ge\frac12$, where $c_\tau=(1-\tau)/\tau=(r-1)^{-1}$, gives the second inequality in (30).

Step 4: the full source space. On the full space G and $\overline{G}$ are injective but need not be invertible. The inequalities hold for $G+\epsilon I$ and $\overline{G}+\epsilon I$, and as $\epsilon\downarrow 0$ the geometric means and parallel sums decrease to those of $G,\overline{G}$, which is how the noninvertible case is defined, while the compressions converge to the invertible $G_F,\overline{G}_F$. The inequalities pass to the limit. □

Appendix D: Complex-shell evaluation and weighted moments

Proof of Lemma 2. The components of an element of $\mathcal{H}_{\mu,U}$ are distributions supported in $\overline{U}$, with $H^{\mp1/2}$ Sobolev norms (Fourier weights $(1+|\mathbf{p}|^2)^{\mp1/2}$) controlled by the covariance norm for fixed μ; their Fourier transforms therefore extend to entire functions on $\mathbb{C}^3$, by the Paley–Wiener–Schwartz theorem [36]. If $\kappa_U(f,g)=0$, then $\hat f(\mathbf{p})+\mathrm{i}\sqrt{\mu^2+|\mathbf{p}|^2}\,\hat g(\mathbf{p})=0$ on $\mathbb{R}^3$; for fixed real $\mathbf{p}_\perp=(p_2,p_3)$, squaring and continuing in the first variable gives $\hat f(z_1,\mathbf{p}_\perp)^2=-(z_1^2+\mu^2+|\mathbf{p}_\perp|^2)\,\hat g(z_1,\mathbf{p}_\perp)^2$ for all $z_1\in\mathbb{C}$. At $z_1=\mathrm{i}\sqrt{\mu^2+|\mathbf{p}_\perp|^2}$ the right side has a simple zero, so it has odd vanishing order unless it vanishes identically, while the left side has even order; both sides vanish, and $f=g=0$. For the density, in the spatial realization $\kappa_U(f,g)=2^{-1/2}\big((\mu^2-\nabla^2)^{-1/4}f+\mathrm{i}(\mu^2-\nabla^2)^{1/4}g\big)$ of the one-particle space as $L^2(\mathbb{R}^3)$, a vector v orthogonal to the complex range gives $u=(\mu^2-\nabla^2)^{-1/4}v$ with $u=0$ and $(\mu^2-\nabla^2)^{1/2}u=0$ on U. The function $W(t)=e^{-t(\mu^2-\nabla^2)^{1/2}}u$, with $t>0$ a Euclidean time, solves the elliptic equation $(-\partial_t^2-\nabla^2+\mu^2)W=0$ with Dirichlet and Neumann data u and $-(\mu^2-\nabla^2)^{1/2}u$ at $t=0$, which vanish on a ball $B\subset U$; its extension by zero across $\{0\}\times B$ is a weak solution on the connected set $((0,\infty)\times\mathbb{R}^3)\cup(\mathbb{R}\times B)$ that vanishes on an open subset, hence everywhere by analytic elliptic regularity and unique continuation [36]. So $u=0$ and $v=0$. For regions with smooth boundary this is the anti-locality argument of Figliolini and Guido [37]. □

The modular operator of a ball.—Let $\mathcal{K}_U=\kappa_U\mathcal{X}_{\mu,U}$ be the closed real subspace of one-particle vectors of real sources in U. By Lemma 2, $\mathcal{K}_U\cap\mathrm{i}\mathcal{K}_U=\{0\}$ and $\mathcal{K}_U+\mathrm{i}\mathcal{K}_U$ is dense, so $\mathsf{s}_U(u+\mathrm{i}v)=u-\mathrm{i}v$ and $\Delta_U=\mathsf{s}_U^*\mathsf{s}_U$ are the standard-subspace operators of Section V, here for $\mathcal{K}=\mathcal{K}_U$. In the polar decomposition $\kappa_U=\mathcal{U}_UG_U^{1/2}$ the factor $\mathcal{U}_U$ is unitary. The defining relation $\mathsf{s}_U\kappa_U\mathsf{f}=\kappa_U\mathsf{c}_U\mathsf{f}$, where c_U is complex conjugation on $\mathcal{H}_{\mu,U}$, and $\overline{G}_U=\mathsf{c}_UG_U\mathsf{c}_U$ give $\|\Delta_U^{1/2}\kappa_U\mathsf{f}\|^2=\|\kappa_U\mathsf{c}_U\mathsf{f}\|^2=\langle\mathsf{f},\overline{G}_U\mathsf{f}\rangle$; comparing quadratic forms on the common form domain $\operatorname{Ran}G_U^{1/2}$ (the set where both quadratic forms are finite),

$$\Delta_U=\mathcal{U}_U\,\overline{G}_UG_U^{-1}\,\mathcal{U}_U^*,\qquad\Delta_U^c\kappa_U=\mathcal{U}_U\,G_U^{1/2-c}\overline{G}_U^{\,c}\tag{D1}$$

for $0\le c\le\frac12$ [28, 38].

The proof of (40) given in Section V applies verbatim to any standard subspace $\mathcal{K}\supset\mathcal{K}_U$, in particular to that of a Lorentz transformed wedge; equivalently, $\Delta_{\mathcal{K}}^c\Delta_U^{-c}$ is a contraction [39, Section 2].

The wedge identity (39).—In rapidity coordinates $p_0=m_\perp\cosh\chi$, $p_1=m_\perp\sinh\chi$, where χ is the rapidity, $\mathbf{p}_\perp=$

(p_2, p_3) and $m_\perp = \sqrt{\mu^2 + |\mathbf{p}_\perp|^2}$, the invariant measure is $\mathrm{d}\mathfrak{m}_\mu = (2(2\pi)^3)^{-1}\,\mathrm{d}\chi\,\mathrm{d}^2p_\perp$ and the boosts along the first axis translate χ. By the Bisognano–Wichmann theorem [29, 30], $\Delta_{\mathcal{W}_0}^{\mathrm{i}y}$ is the boost of rapidity $2\pi y$, so after a Fourier transform in χ it is multiplication by $e^{-2\pi\mathrm{i}yk}$, k the variable dual to χ, and $\Delta_{\mathcal{W}_0}^{c}$ is multiplication by $e^{-\varphi k}$, $\varphi = 2\pi c$. By the Paley–Wiener theorem for a strip [36], a wave function ψ whose restriction to each line of fixed $\mathbf{p}_\perp$ extends holomorphically to $0 < \operatorname{Im}\chi < \varphi$, with square-integrable bounds, lies in the domain of $\Delta_{\mathcal{W}_0}^{c}$ and $(\Delta_{\mathcal{W}_0}^{c}\psi)(\chi, \mathbf{p}_\perp) = \psi(\chi + \mathrm{i}\varphi, \mathbf{p}_\perp)$. For a smooth source f with compact support in $\{x^1 < 0\}$, the function $\chi \mapsto \Psi_{\mathsf{f}}(\mathcal{L}_1(\chi)(m_\perp, 0, \mathbf{p}_\perp))$, where $\mathcal{L}_1(\chi)$ is the boost of rapidity χ along the first axis, $(p_0, p_1) \mapsto (p_0\cosh\chi + p_1\sinh\chi,\ p_1\cosh\chi + p_0\sinh\chi)$, is entire and equals $\kappa_U\mathsf{f}$ at real χ. On the strip, write the continued rapidity as $\chi + \mathrm{i}\varphi'$ with $0 \le \varphi' \le \varphi$; then $|e^{-\mathrm{i}\mathbf{z}\cdot\mathbf{x}}| \le 1$ on the support because $\operatorname{Im} z_1 = m_\perp\cosh\chi\sin\varphi' \ge 0$, and integration by parts gives rapid decay in $|\mathbf{z}|$. Hence $\Delta_{\mathcal{W}_0}^{c}\kappa_U\mathsf{f} = \Psi_{\mathsf{f}} \circ b_\varphi$, since $\mathcal{L}_1(\chi + \mathrm{i}\varphi) = \mathcal{L}_1(\mathrm{i}\varphi)\mathcal{L}_1(\chi)$. Translating the wedge edge by ℓ conjugates Δ by the translation $e^{-\mathrm{i}\ell p_1}$, and the continuation $p_1 \mapsto p_1\cos\varphi + \mathrm{i}p_0\sin\varphi$ turns $e^{\mathrm{i}\ell p_1}$ into a phase times $e^{-\ell p_0\sin\varphi}$, which gives (39) for smooth sources in the base of $\mathcal{W}_\ell$. For a proper orthochronous Lorentz transformation $\mathcal{L}$, the subspace of $\mathcal{L}\mathcal{W}_\ell$ is $\mathsf{U}(\mathcal{L})\mathcal{K}_{\mathcal{W}_\ell}$, where U is the unitary one-particle representation of the Poincaré group, $(\mathsf{U}(\mathcal{L})\psi)(p) = \psi(\mathcal{L}^{-1}p)$; its modular operator is the conjugate of $\Delta_{\mathcal{W}_\ell}$, and if $D_T \subset \mathcal{L}\mathcal{W}_\ell$ then every real source in $U \subset B_T(0)$ has $\kappa_U\mathsf{f} \in \mathcal{K}_{\mathcal{L}\mathcal{W}_\ell}$—and so, by complex linearity, every $\mathsf{f} \in \mathcal{H}_{\mu,U}$ has $\kappa_U\mathsf{f} \in \mathcal{K}_{\mathcal{L}\mathcal{W}_\ell} + \mathrm{i}\mathcal{K}_{\mathcal{L}\mathcal{W}_\ell} = \operatorname{Dom}\mathsf{s}_{\mathcal{L}\mathcal{W}_\ell}$—with $\Psi_{\mathsf{f}}(\mathcal{L}b_\varphi(p))$ in place of $\Psi_{\mathsf{f}}(b_\varphi(p))$ by Lorentz covariance of the field and the identity theorem on the connected complex manifold $H_\mu^{\mathbb{C}}$. Finally, since $\mathsf{s}_{\mathcal{W}_\ell}$ fixes $\mathcal{K}_{\mathcal{W}_\ell}$ pointwise, $\|\Delta_{\mathcal{W}_\ell}^{1/2}\xi\| = \|\xi\|$ there, so $\|\Delta_{\mathcal{W}_\ell}^{c}\xi\| \le 2\|\xi\|$, and approximation of an arbitrary $\mathsf{f} \in \mathcal{H}_{\mu,U}$ by smooth sources, together with the continuity of $\mathsf{f} \mapsto \Psi_{\mathsf{f}}(z)$ on $\mathcal{H}_{\mu,U}$ (the values $\hat f(\mathbf{z})$, $\hat g(\mathbf{z})$ are pairings of $f \in H^{-1/2}$ and $g \in H^{1/2}$ with a fixed smooth function), extends the identity to all sources.

Averaged wedges.—For $U \subset B_T(0)$ and a compact family $\mathscr{L}$ of Lorentz transformations, the choice $\ell_{\mathscr{L}} > \sqrt{2}\sup_{\mathcal{L}\in\mathscr{L}}\|\mathcal{L}^{-1}\|_{\mathrm{E}}$, with $\|\cdot\|_{\mathrm{E}}$ the Euclidean operator norm on $\mathbb{R}^4$, ensures $D_T \subset \mathcal{L}\mathcal{W}_{\ell_{\mathscr{L}}T}$ for every $\mathcal{L} \in \mathscr{L}$, because D_T lies in the Euclidean ball of radius T and $x^1 + |x^0| \le \sqrt{2}|x|_{\mathrm{E}}$. The wedge identity and (40), integrated against a probability measure ϑ on $\mathscr{L}$, give

$$\int_{\mathscr{L}}\int_{H_\mu^+} e^{-2\ell_{\mathscr{L}}Tp_0\sin\varphi}|\Psi_{\mathsf{f}}(\mathcal{L}b_\varphi(p))|^2\,\mathrm{d}\mathfrak{m}_\mu(p)\,\mathrm{d}\vartheta(\mathcal{L}) \le \|G_U^{1/2-c}\overline{G}_U^{c}\mathsf{f}\|^2. \tag{D2}$$

Lemma 7 (Regular shell patches)**.** *Let $0 < c < \frac12$, $\varphi = 2\pi c$, and let $\mathscr{Z} = \{(t,z) : 0 \le t \le 1,\ z \in H_t^{\mathbb{C}},\ |z|_{\mathrm{E}} \le 16,\ (\operatorname{Re} z)^2 \ge 16\}$, where $(\operatorname{Re} z)^2 = \operatorname{Re} z\cdot\operatorname{Re} z$ is the Minkowski square, positive for spacelike vectors. There are a compact set $\mathscr{L}_c$ of proper orthochronous Lorentz transformations with $\sup_{\mathscr{L}_c}\|\mathcal{L}^{-1}\|_{\mathrm{E}} < 32/\sin\varphi$, a probability measure ϑ_c on $\mathscr{L}_c$ with smooth density with respect to Haar measure, a neighbourhood $\mathcal{V}$ of $\mathscr{Z}$ in the shell family $\{(t,z) : 0 \le t \le 1,\ z \in H_t^{\mathbb{C}}\}$, and $\delta_c > 0$, such that the image of the measure $\mathrm{d}\vartheta_c \otimes \mathbf{1}_{\{p_0\le M_c\}}\,\mathrm{d}\mathfrak{m}_t$, $M_c = 32/\sin\varphi$, under $(\mathcal{L}, p) \mapsto \mathcal{L}b_\varphi(p)$ assigns to every subset of the slice $\mathcal{V}_t = \{z : (t,z) \in \mathcal{V}\}$ at least δ_c times its six-dimensional Euclidean volume* vol*; the formulas for $\mathrm{d}\mathfrak{m}_t$ extend to $t = 0$, the measure on the forward light cone. Moreover, there are a chart radius $r_c > 0$ and $C_{\mathrm{sub}} \ge 1$ such that every $(t,z) \in \mathscr{Z}$ has a holomorphic chart of radius r_c in $H_t^{\mathbb{C}}$ centred at z, with image $\mathcal{B}_z \subset \mathcal{V}_t$ and volume Jacobian between 1 and C_{sub}; hence every nonnegative plurisubharmonic F on $\mathbb{C}^4$ (subharmonic on every complex line), in particular the squared modulus of any holomorphic function, satisfies $F(z) \le C_{\mathrm{sub}}\,\mathrm{vol}(\mathcal{B}_z)^{-1}\int_{\mathcal{B}_z} F\,\mathrm{d}\,\mathrm{vol}$.*

Proof (condensed). A point $z = u + \mathrm{i}v \in H_t^{\mathbb{C}}$ with $(t,z) \in \mathscr{Z}$ has $u\cdot v = 0$, $u^2 = \rho_0^2$ with $\rho_0 \in [4, 16]$, and $v^2 = t^2 + \rho_0^2$, since $z\cdot z = u^2 - v^2 + 2\mathrm{i}\,u\cdot v = -t^2$. The pair (u, v) is the Lorentz image of the real and imaginary parts of $b_\varphi(p)$ at $p_1 = p_3 = 0$, $p_0 = \sqrt{t^2 + \rho_0^2}/\sin\varphi$, $p_2 = \sqrt{p_0^2 - t^2}$, where these parts are orthogonal with the same Minkowski squares; the matching transformation $\mathcal{L} = \mathcal{L}_{\mathrm{tar}}\mathcal{L}_{\mathrm{src}}^{-1}$, where $\mathcal{L}_{\mathrm{src}}$ and $\mathcal{L}_{\mathrm{tar}}$ are proper orthochronous transformations carrying the standard frame to orthonormal frames whose first two spacelike legs are the two pairs, satisfies $\|\mathcal{L}\|_{\mathrm{E}} = \|\mathcal{L}^{-1}\|_{\mathrm{E}} \le 2\sqrt{34}/\sin\varphi < 32/\sin\varphi$ (the Euclidean norms of a Lorentz matrix and of its inverse coincide) and $p_0 \le \sqrt{257}/\sin\varphi < 32/\sin\varphi$: for the normalized legs u/ρ_0 and $v/\sqrt{t^2 + \rho_0^2}$, the bound $|z|_{\mathrm{E}}^2 = 2(u_0^2 + v_0^2) + 2\rho_0^2 + t^2 \le 256$ gives

$$\frac{u_0^2}{\rho_0^2} + \frac{v_0^2}{t^2 + \rho_0^2} \le \frac{u_0^2 + v_0^2}{\rho_0^2} \le \frac{128}{\rho_0^2} - 1 \le 7,$$

so, completing the pair by a third spacelike leg with zero time component, the Lorentz factor of $\mathcal{L}_{\mathrm{tar}}$ (the time component of the image of the time axis) is at most $\sqrt{8}$; that of $\mathcal{L}_{\mathrm{src}}$ is $\sqrt{1 + (17/16)\cot^2\varphi} \le \sqrt{17}/(4\sin\varphi)$, since $t \le 1$ and $\rho_0^2 \ge 16$; a proper orthochronous matrix of Lorentz factor γ has Euclidean norm $\gamma + \sqrt{\gamma^2 - 1} \le 2\gamma$, and the two norms multiply. The map $(\mathcal{L}, p) \mapsto \mathcal{L}b_\varphi(p)$ is a submersion at such preimages: the Lorentz orbit of the ordered pair (u, v) has dimension five (the stabilizer of an orthogonal spacelike pair is the one-parameter boost group of the complementary timelike plane), and varying p_2 changes the invariant $(\operatorname{Re} z)^2$ with nonzero derivative:

$$(\operatorname{Re} b_\varphi)^2 = -p_0^2\cos^2\varphi + p_2^2 = (t^2 + p_2^2)\sin^2\varphi - t^2,$$
$$\partial_{p_2}(\operatorname{Re} b_\varphi)^2 = 2p_2\sin^2\varphi \ne 0.$$

Including t as a parameter, the submersion theorem and Fubini's theorem give, near each point of $\mathscr{Z}$, a push-forward density that is continuous in (t,z), $t = 0$ included, and positive (the preimages have $p_2 \ge \rho_0 \ge 4$, uniformly away from the apex of the cone, so $\mathrm{d}\mathfrak{m}_t$ and the local Jacobians stay regular down to $t = 0$); compactness of $\mathscr{Z}$ gives a finite cover, one

smooth Lorentz density supported in the open set $\{\|\mathcal{L}^{-1}\|_{\mathrm{E}} < 32/\sin\varphi\}$, and $\delta_c > 0$. The charts come from the holomorphic implicit function theorem applied to $z \cdot z = -t^2$ in a coordinate z_k with $|z_k| \geq 4/\sqrt{3}$, uniformly over the compact set $\mathscr{Z}$; the Jacobian bounds and the sub-mean-value property of F composed with the chart give the last claim. □

Lemma 8 (Null analytic discs)**.** *If $z \in H_\mu^{\mathbb{C}}$, $\mathcal{E} \geq \mu$ and $|z|_{\mathrm{E}} \leq 2\mathcal{E}$, there is an analytic disc $\zeta \mapsto z(\zeta)$ in $H_\mu^{\mathbb{C}}$, $|\zeta| \leq 1$, with $z(0) = z$, whose boundary points satisfy $|\mathrm{Re}\,\mathbf{z}(\zeta)| \geq 6\mathcal{E}$, $|\mathrm{Re}\,z_0(\zeta)| \leq 2\mathcal{E}$, $(\mathrm{Re}\,z(\zeta))^2 \geq 32\mathcal{E}^2$ and $|z(\zeta)|_{\mathrm{E}} < 14\mathcal{E}$; in particular $(\mu/\mathcal{E}, z(\zeta)/\mathcal{E}) \in \mathscr{Z}$.*

Proof. Choose $\mathbf{n} \in \mathbb{C}^3$, $\mathbf{n} \neq 0$, with $\mathbf{z} \cdot \mathbf{n} = 0$ and $\mathbf{n} \cdot \mathbf{n} = 0$ (an *isotropic* vector): the first equation defines a complex subspace of dimension at least two, and a quadratic form on such a subspace has a nonzero isotropic vector. Normalizing, $|\mathrm{Re}\,\mathbf{n}| = |\mathrm{Im}\,\mathbf{n}| = 1$ and $\mathrm{Re}\,\mathbf{n} \perp \mathrm{Im}\,\mathbf{n}$, so $|\mathrm{Re}(\zeta\mathbf{n})| = 1$ for $|\zeta| = 1$. The disc $z(\zeta) = z + 8\mathcal{E}\zeta(0, \mathbf{n})$ lies in $H_\mu^{\mathbb{C}}$ because $\mathbf{z} \cdot \mathbf{n} = \mathbf{n} \cdot \mathbf{n} = 0$, and on its boundary z_0 is unchanged while $|\mathrm{Re}\,\mathbf{z}(\zeta)| \geq 8\mathcal{E} - |\mathrm{Re}\,\mathbf{z}| \geq 6\mathcal{E}$; the Minkowski-square bound follows, and $|z(\zeta)|_{\mathrm{E}} \leq (2 + 8\sqrt{2})\mathcal{E} < 14\mathcal{E}$. □

Proof of Theorem 6. Let f be smooth with compact support in U, and write $\mu = t\mathcal{E}$ with $t \in (0, 1]$; the substitution $p = \mathcal{E}p'$ scales the shell measure by $\mathcal{E}^2$. For $(t, z) \in \mathscr{Z}$, the plurisubharmonic function $F = |\Psi_{\mathsf{f}}(\mathcal{E}\,\cdot)|^2$, the chart bound of Lemma 7, the push-forward bound on $\mathcal{B}_z \subset \mathcal{V}_t$, the insertion of the factor $e^{2\ell_c T(\mathcal{E}M_c - p_0)\sin\varphi} \geq 1$ on $\{p_0 \leq \mathcal{E}M_c\}$, and (D2) with $\mathscr{L} = \mathscr{L}_c$, $\ell_{\mathscr{L}} = \ell_c = 64/\sin\varphi$ (which exceeds $\sqrt{2} \cdot 32/\sin\varphi$, as required) give

$$|\Psi_{\mathsf{f}}(\mathcal{E}z)|^2 \leq \frac{C_{\mathrm{sub}}\mathcal{E}^{-2}e^{2\ell_c M_c T\mathcal{E}\sin\varphi}}{\delta_c\,\mathrm{vol}(\mathcal{B}_z)}\|G_U^{1/2-c}\overline{G}_U^{c}\mathsf{f}\|^2,$$

with $\ell_c M_c \sin\varphi = 2048/\sin(2\pi c) = \lambda_c$ and $C_c = \max\{1, C_{\mathrm{sub}}/(\delta_c \inf_{\mathscr{Z}}\mathrm{vol}(\mathcal{B}_z))\}$, finite because $\mathrm{vol}(\mathcal{B}_z) \geq \mathrm{vol}(B_{r_c})$. For a general z with $|z|_{\mathrm{E}} \leq 2\mathcal{E}$, Lemma 8 and the mean value property of holomorphic functions, $|\Psi_{\mathsf{f}}(z)| \leq \max_{|\zeta|=1}|\Psi_{\mathsf{f}}(z(\zeta))|$, reduce the claim to boundary points of the form $\mathcal{E}z'$ with $(t, z') \in \mathscr{Z}$. Density of smooth sources and continuity of both sides in f give the statement on $\mathcal{H}_{\mu,U}$. The set U enters only through $U \subset B_T(0)$. □

Weighted moments.—For a multi-index $\alpha \in \mathbb{N}^3$ and a centre $\mathbf{x}_0$, let $\mathsf{m}_\alpha^f(\mathsf{f}) = \int(\mathbf{x} - \mathbf{x}_0)^\alpha f(\mathbf{x})\,\mathrm{d}^3x$ and $\mathsf{m}_\alpha^g(\mathsf{f}) = \int(\mathbf{x} - \mathbf{x}_0)^\alpha g(\mathbf{x})\,\mathrm{d}^3x$. For $\frac{1}{2} \leq b < 1$ fix

$$\Lambda_b = \frac{2^{15}}{\sin(\pi b)} = 16\lambda_{b/2}, \qquad B_b = 5\Lambda_b. \tag{D3}$$

Lemma 9 (Weighted moments)**.** *Let $\frac{1}{2} \leq b < 1$. There is a finite M_b, depending only on b, such that for all $\mu, T > 0$, all $\mathbf{x}_0 \in \mathbb{R}^3$, every nonempty bounded open $U \subset B_T(\mathbf{x}_0)$, every $\mathsf{f} \in \mathcal{H}_{\mu,U}$ and every $\alpha \in \mathbb{N}^3$,*

$$\begin{aligned}|\mathsf{m}_\alpha^f(\mathsf{f})| &\leq M_b\frac{T}{1+\mu T}e^{\Lambda_b\mu T}(B_bT)^{|\alpha|}\|\mathsf{R}_U^b\mathsf{f}\|,\\ |\mathsf{m}_\alpha^g(\mathsf{f})| &\leq M_b\frac{T^2}{(1+\mu T)^2}e^{\Lambda_b\mu T}(B_bT)^{|\alpha|}\|\mathsf{R}_U^b\mathsf{f}\|.\end{aligned} \tag{D4}$$

By the Riesz representation theorem, each bound says that there is a vector $\mathsf{u} \in \mathcal{H}_{\mu,U}$, of norm at most the stated constant, with $\mathsf{m}_\alpha^\bullet(\mathsf{f}) = \langle \mathsf{u}, \mathsf{R}_U^b\mathsf{f}\rangle$ for all f.

Proof of Lemma 9. Work in coordinates centred at $\mathbf{x}_0$; translation preserves ς, v_μ and R_U. By (38), for $\mathcal{E} \geq \mu$, $|\Psi_{\mathsf{f}}(z)| \leq \big(C_{(1-b)/2}^{1/2} + C_{b/2}^{1/2}\big)\mathcal{E}^{-1}e^{\lambda_{b/2}T\mathcal{E}}\|\mathsf{R}_U^b\mathsf{f}\|$ on $\{z \in H_\mu^{\mathbb{C}} : |z|_{\mathrm{E}} \leq 2\mathcal{E}\}$. For $\mathbf{z} \in \mathbb{C}^3$ take $\mathcal{E} = \mu + 1/(\Lambda_bT) + |\mathbf{z}|$. The two shell points $z_\pm = (\pm\sqrt{\mu^2 + \mathbf{z}\cdot\mathbf{z}}, \mathbf{z})$ have $|z_\pm|_{\mathrm{E}} < \sqrt{2}\mathcal{E}$, and $\hat{f}(\mathbf{z}) = \frac{1}{2}(\Psi_{\mathsf{f}}(z_+) + \Psi_{\mathsf{f}}(z_-))$, so $|\hat{f}(\mathbf{z})| \leq \big(C_{(1-b)/2}^{1/2} + C_{b/2}^{1/2}\big)\mathcal{E}^{-1}e^{\lambda_{b/2}T\mathcal{E}}\|\mathsf{R}_U^b\mathsf{f}\|$. For $\hat{g}$, the difference $\Psi_{\mathsf{f}}(z_+) - \Psi_{\mathsf{f}}(z_-) = 2\mathrm{i}\sqrt{\mu^2 + \mathbf{z}\cdot\mathbf{z}}\,\hat{g}(\mathbf{z})$ is useless near zero energy, so move z_1 on a circle: for $|\zeta| = 1$ put $\mathbf{z}_\zeta = (z_1 + 8\mathcal{E}\zeta, z_2, z_3)$. Then $|\mu^2 + \mathbf{z}_\zeta \cdot \mathbf{z}_\zeta| \geq 47\mathcal{E}^2$, the shell points over $\mathbf{z}_\zeta$ have Euclidean norm below $14\mathcal{E}$, and (38) at the energy $7\mathcal{E}$ gives $|\hat{g}(\mathbf{z}_\zeta)| \leq \big(C_{(1-b)/2}^{1/2} + C_{b/2}^{1/2}\big)(7\sqrt{47})^{-1}\mathcal{E}^{-2}e^{7\lambda_{b/2}T\mathcal{E}}\|\mathsf{R}_U^b\mathsf{f}\|$; the mean value property of $\zeta \mapsto \hat{g}(\mathbf{z}_\zeta)$ transfers the bound to $\hat{g}(\mathbf{z})$. Since $\Lambda_b = 16\lambda_{b/2}$, both exponents are at most $\Lambda_bT\mathcal{E} = \Lambda_b\mu T + 1 + \Lambda_bT|\mathbf{z}|$, and $\mathcal{E}^{-1} \leq \Lambda_bT/(1 + \mu T)$; so, for all $\mathbf{z} \in \mathbb{C}^3$,

$$|\hat{f}(\mathbf{z})| \leq e\big(C_{(1-b)/2}^{1/2} + C_{b/2}^{1/2}\big)\Lambda_b\frac{T}{1+\mu T}\,e^{\Lambda_bT(\mu+|\mathbf{z}|)}\|\mathsf{R}_U^b\mathsf{f}\|,$$

$$|\hat{g}(\mathbf{z})| \leq e\big(C_{(1-b)/2}^{1/2} + C_{b/2}^{1/2}\big)\Lambda_b^2\frac{T^2}{(1+\mu T)^2}\,e^{\Lambda_bT(\mu+|\mathbf{z}|)}\|\mathsf{R}_U^b\mathsf{f}\|.$$

Finally $\partial^\alpha\hat{f}(0) = (-\mathrm{i})^{|\alpha|}\mathsf{m}_\alpha^f(\mathsf{f})$, and for $|\alpha| \geq 1$ the Cauchy estimate on the polydisc of radius $|\alpha|/(\sqrt{3}\Lambda_bT)$, where $e^{\Lambda_bT|\mathbf{z}|} \leq e^{|\alpha|}$, gives the factor $\alpha!(\sqrt{3}\Lambda_bT/|\alpha|)^{|\alpha|}e^{|\alpha|} \leq (e\sqrt{3}\Lambda_bT)^{|\alpha|} \leq (B_bT)^{|\alpha|}$; for $\alpha = 0$ the displayed bounds at $\mathbf{z} = 0$ are (D4) directly. Thus (D4) holds with $M_b = e\big(C_{(1-b)/2}^{1/2} + C_{b/2}^{1/2}\big)\Lambda_b^2$. □

Appendix E: The forward and reverse estimates

Throughout this appendix, C is a universal constant whose value may change from occurrence to occurrence; subscripted constants keep their values.

Covariance kernels.—Away from the diagonal, the covariance (6) is given by integral kernels: for sources with disjoint supports, $\mathsf{v}_\mu(\mathsf{f}, \mathsf{f}') = \iint[f(\mathbf{x})\mathfrak{X}_\mu(\mathbf{y} - \mathbf{x})f'(\mathbf{y}) + g(\mathbf{x})\mathfrak{P}_\mu(\mathbf{y} - \mathbf{x})g'(\mathbf{y})]\,\mathrm{d}^3x\,\mathrm{d}^3y$ with

$$\mathfrak{X}_\mu(\mathbf{w}) = \frac{\mu}{4\pi^2\varrho}K_1(\mu\varrho), \qquad \mathfrak{P}_\mu(\mathbf{w}) = -\frac{\mu^2}{4\pi^2\varrho^2}K_2(\mu\varrho), \tag{E1}$$

where $\varrho = \sqrt{\mathbf{w}\cdot\mathbf{w}}$ and K_1, K_2 are modified Bessel functions of the second kind [40]. These formulas come from the heat-kernel representation of $(\mu^2 - \nabla^2)^{-1/2}$ and the integral representation $K_1(x) = (x/4)\int_0^\infty t^{-2}e^{-t-x^2/(4t)}\,\mathrm{d}t$ [40, Eq. 10.32.10], and $\mathfrak{P}_\mu = (-\nabla^2 + \mu^2)\mathfrak{X}_\mu$ by the Bessel recurrences. From $K_j(x) =$

$\int_0^\infty e^{-x\cosh u}\cosh(ju)\,\mathrm{d}u$ [40, Eq. 10.32.9], for $x>0$,

$$xK_1(x)\le 3(1+x)^{1/2}e^{-x},\qquad x^2K_2(x)\le 8(1+x)^{3/2}e^{-x}, \tag{E2}$$

and $|K_j(z)|\le K_j(\operatorname{Re} z)$ for $\operatorname{Re} z>0$.

Proof of Theorem 4. Measure $\mathbf{x}$ and $\mathbf{y}$ from the centres of the two balls, so that $\mathbf{w}=s\mathbf{e}_1+\mathbf{y}-\mathbf{x}$, and put $d_{0,b}=4\sqrt{3}\,B_b$ and $S_b=2(2\Lambda_b+2d_{0,b})=(4+80\sqrt{3})\,\Lambda_b$, which is at least 4 and $4d_{0,b}$; let $s\ge S_b$ and $\xi=\mu s$.

Step 1: the kernels on a polydisc. On the six-variable polydisc (a product of discs, one per complex coordinate) $|x_j|,|y_j|\le 2B_b$ one has $\mathbf{w}\cdot\mathbf{w}=s^2(1+\epsilon)$ with $|\epsilon|\le 2d_{0,b}/s+(d_{0,b}/s)^2\le 9/16$ since $s\ge 4d_{0,b}$, so the principal branch $\varrho=s\sqrt{1+\epsilon}$ is holomorphic there with $\operatorname{Re}\varrho\ge s-2d_{0,b}\ge s/2$ and $|\varrho|\le 5s/4$. With (E2), the polydisc suprema of the two kernels are at most $Cs^{-2}(1+\xi)^{1/2}e^{-\xi+2\mu d_{0,b}}$ and $Cs^{-4}(1+\xi)^{3/2}e^{-\xi+2\mu d_{0,b}}$, and their Taylor coefficients $c^f_{\alpha\beta},c^g_{\alpha\beta}$ in $\mathbf{x}^\alpha\mathbf{y}^\beta$ are bounded by these suprema times $(2B_b)^{-|\alpha|-|\beta|}$. The series converge absolutely on the unit balls.

Step 2: the rank-one series. For smooth real sources f,f' in the two balls, term-by-term integration gives

$$\mathsf{v}_\mu(\mathsf{f},\mathsf{f}')=\sum_{\alpha,\beta}\bigl(c^f_{\alpha\beta}\mathsf{m}^f_\alpha(\mathsf{f})\mathsf{m}^f_\beta(\mathsf{f}')+c^g_{\alpha\beta}\mathsf{m}^g_\alpha(\mathsf{f})\mathsf{m}^g_\beta(\mathsf{f}')\bigr).$$

By Lemma 9 at $T=1$ and the Riesz representation, there are, for $\bullet\in\{f,g\}$, vectors $\mathsf{u}^\bullet_{A,\alpha},\mathsf{u}^\bullet_{B,\beta}$ with $\mathsf{m}^\bullet_\alpha(\mathsf{f})=\langle\mathsf{u}^\bullet_{A,\alpha},\mathsf{R}^b_A\mathsf{f}\rangle$, $\mathsf{m}^\bullet_\beta(\mathsf{f}')=\langle\mathsf{u}^\bullet_{B,\beta},\mathsf{R}^b_B\mathsf{f}'\rangle$, of norms at most $M_be^{\Lambda_b\mu}B_b^{|\alpha|}/(1+\mu)$ for $\bullet=f$ and $M_be^{\Lambda_b\mu}B_b^{|\alpha|}/(1+\mu)^2$ for $\bullet=g$. Put $X_{b,\mu,s}=\sum_{\bullet,\alpha,\beta}c^\bullet_{\alpha\beta}\,\mathsf{u}^\bullet_{A,\alpha}\langle\mathsf{u}^\bullet_{B,\beta},\cdot\rangle$. Since $B_b/(2B_b)=\frac12$ and $\sum_{\alpha,\beta}2^{-|\alpha|-|\beta|}=64$, the series converges in trace norm, with

$$\|X_{b,\mu,s}\|_1\le CM_b^2e^{(2\Lambda_b+2d_{0,b})\mu-\xi}\times\Bigl[\frac{s^{-2}(1+\xi)^{1/2}}{(1+\mu)^2}+\frac{s^{-4}(1+\xi)^{3/2}}{(1+\mu)^4}\Bigr].$$

Step 3: the constant and the identification. Since $s\ge S_b=2(2\Lambda_b+2d_{0,b})$, one has $(2\Lambda_b+2d_{0,b})\mu\le\xi/2$, so the right side is at most $2CM_b^2s^{-2}(1+\xi)^{3/2}e^{-\xi/2}\le F_bs^{-2}$ with $F_b=2\cdot3^{3/2}e^{-1}CM_b^2$, the supremum of $(1+\xi)^{3/2}e^{-\xi/2}$ being attained at $\xi=2$. On pairs of smooth real sources, $\langle\mathsf{f},\mathsf{R}^b_AX_{b,\mu,s}\mathsf{R}^b_B\mathsf{f}'\rangle$ equals the series of Step 2, hence $\mathsf{v}_\mu(\mathsf{f},\mathsf{f}')=\langle\mathsf{f},\mathsf{C}_{\mu,s}\mathsf{f}'\rangle$; by sesquilinearity and density the two bounded operators coincide. □

Smooth kernels in the joint norm.—For $s\ge5$, let K be a kernel on $(B_2(0)\cup B_2(s\mathbf{e}_1))^2$ with derivatives of order at most two in each variable bounded by M_0. Fix a smooth function equal to 1 on U_s, supported within distance 1 of it and with s-independent derivative bounds, and let K° be K multiplied by that function in each variable. The Hilbert–Schmidt norm (the square root of the sum of the squared singular values) of $(1-\nabla^2)^{1/4}K^\circ(1-\nabla^2)^{1/4}$ is bounded by CM_0, and the $L^2(\mathbb{R}^3)$ comparisons $\|(1-\nabla^2)^{-1/4}f\|_{L^2}\le\max\{1,\mu\}^{1/2}\|(\mu^2-\nabla^2)^{-1/4}f\|_{L^2}$ and, for g supported in U_s, $\|g\|^2_{L^2}\le2\|(\mu^2-\nabla^2)^{1/4}g\|^2_{L^2}$ (less than half of $\|g\|^2_{L^2}$ sits at frequencies $|\mathbf{p}|<1$, by the volume of U_s) give, for sources $\mathsf{f}=(f,g)$, $\mathsf{f}'=(f',g')$ supported in U_s and in the joint norm,

$$\Bigl|\iint fKf'\Bigr|\le C\max\{1,\mu\}M_0\|\mathsf{f}\|\|\mathsf{f}'\|,\qquad\Bigl|\iint gKg'\Bigr|\le CM_0\|\mathsf{f}\|\|\mathsf{f}'\|, \tag{E3}$$

with C independent of $s\ge5$ and μ.

The polynomial *selectors*

$$\mathsf{p}_N(z)=(1-z)^N\sum_{k=0}^{N-1}\binom{N+k-1}{k}z^k \tag{E4}$$

have degree $2N-1$; p_N-1 vanishes to order N at $z=0$ and p_N vanishes to order N at $z=1$. So $\mathsf{p}_N(x_1/s)\approx1$ near the ball $B_1(0)$ and $\mathsf{p}_N(x_1/s)\approx0$ near $B_1(s\mathbf{e}_1)$: multiplying a kernel by such factors switches it on between the two balls and off within each ball, as the kernel of $\mathsf{D}_{\mu,s}$ requires.

Lemma 10 (Polynomial increments)**.** *There are universal finite constants $C_1,C_2,c_{\mathrm{pol}},K_{\mathrm{pol}}$ and S_{pol} such that for every $\mu>0$ and $s\ge S_{\mathrm{pol}}$, with $\xi=\mu s$, there are real self-adjoint finite-rank operators F_n on $\mathcal{H}_{\mu,U_s}$ with polynomial kernels, $\mathsf{F}_0=0$ and $\mathsf{F}_n\to\mathsf{D}_{\mu,s}$ in operator norm, whose increments $\mathsf{G}_n=\mathsf{F}_{n+1}-\mathsf{F}_n$ satisfy*

$$\|\mathsf{G}_n\|_1\le C_1(n+1)^3s^{-2}(1+\xi)^2e^{-\xi/2}(c_{\mathrm{pol}}/s)^n, \tag{E5}$$

$$\mathsf{G}_n=\mathsf{R}^{1/2}_{U_s}\mathsf{B}_n\mathsf{R}^{1/2}_{U_s},\quad\|\mathsf{B}_n\|_1\le C_2e^{-\xi/2}e^{2\Lambda_0\mu T}K^n_{\mathrm{pol}}, \tag{E6}$$

where $\Lambda_0=\Lambda_{1/2}=2^{15}$ and, here and below, $B_{1/2},M_{1/2}$ are the constants of Lemma 9 at $b=\frac12$, $T=s/2+1$ is the radius of a ball about $s\mathbf{e}_1/2$ containing U_s (the value at which Lemma 9 is applied), and the B_n are real self-adjoint of finite rank.

Proof (condensed). By (34) and (E1), the kernel of $\mathsf{D}_{\mu,s}$ is $-\mathfrak{X}_\mu(\mathbf{y}-\mathbf{x})$ on field sources and $-\mathfrak{P}_\mu(\mathbf{y}-\mathbf{x})$ on momentum sources when $\mathbf{x}$ and $\mathbf{y}$ lie in different balls, and 0 when they lie in the same ball. The approximating kernel $q_{n,\bullet}$ below is a Taylor polynomial of the cross kernel, multiplied by selectors that keep it only between the balls. In the coordinates $\mathbf{u}=\mathbf{x}/s$, $\mathbf{v}=\mathbf{y}/s$, with components u_j,v_j, the cross kernels $\mathfrak{X}_\mu(s(\mathbf{v}-\mathbf{u}))$ and $\mathfrak{P}_\mu(s(\mathbf{v}-\mathbf{u}))$ are holomorphic on the polydisc of radius $2r_0=1/50$ (so $r_0=1/100$) about $(0,\mathbf{e}_1)$, with suprema $M_f=Cs^{-2}(1+\xi)^{1/2}e^{-\xi/2}$ and $M_g=Cs^{-4}(1+\xi)^{3/2}e^{-\xi/2}$ by (E2), and Taylor coefficients bounded by $M_\bullet r_0^{-|\alpha|-|\beta|}$. Let $\mathcal{T}_{n,\bullet}$ be their Taylor polynomials of degree $n-1$ in the six variables $(\mathbf{u},\mathbf{v}-\mathbf{e}_1)$ about the origin, and define the polynomial kernel

$$\begin{aligned}q_{n,\bullet}(\mathbf{x},\mathbf{y})=\;&-\mathsf{p}_{2n}(u_1)\mathsf{p}_{2n}(1-v_1)\mathcal{T}_{n,\bullet}(\mathbf{u},\mathbf{v}-\mathbf{e}_1)\\&-\mathsf{p}_{2n}(v_1)\mathsf{p}_{2n}(1-u_1)\mathcal{T}_{n,\bullet}(\mathbf{v},\mathbf{u}-\mathbf{e}_1),\end{aligned}$$

real and symmetric, of degree at most $9n$, with the selectors (E4); F_n is the operator with the field-source form of $q_{n,f}$ plus the momentum-source form of $q_{n,g}$. Its form depends

on f only through the moments of degree at most $9n$, so $\operatorname{rank}\mathsf{F}_n \le C(n+1)^3$. The endpoint bounds $|\mathsf{p}_N(z)-1| \le (4r(1+r))^N/(2(1-2r))$ for $|z| \le r$ and $|\mathsf{p}_N(z)| \le (4r(1+r))^N$ for $|z-1| \le r$, $r < \frac{1}{2}$, together with the Taylor tail in six variables,

$$\sum_{k\ge n}\binom{k+5}{5}\Big(\frac{r}{r_0}\Big)^k \le C(n+1)^5\Big(\frac{r}{r_0}\Big)^n, \qquad r \le \frac{r_0}{2},$$

applied at $r = 3/s$, show that on the radius-$3/s$ complex neighbourhoods of the four blocks $B_2(\cdot)\times B_2(\cdot)$ of $(B_2(0)\cup B_2(s\mathbf{e}_1))^2$, the target kernel $\mathfrak{K}_\bullet$ of $\mathsf{D}_{\mu,s}$, equal to $-\mathfrak{X}_\mu$ (for $\bullet = f$) and $-\mathfrak{P}_\mu$ (for $\bullet = g$) on the two cross blocks $B_2(0)\times B_2(s\mathbf{e}_1)$ and $B_2(s\mathbf{e}_1)\times B_2(0)$, and to zero on the two diagonal blocks, satisfies $\sup|\mathfrak{K}_\bullet - q_{n,\bullet}| \le C(n+1)^6 M_\bullet (c_0/s)^n$ with a universal $c_0 \ge 3/r_0$. The four blocks separate as follows: on its own cross block, a summand of $q_{n,\bullet}$ differs from the target by the Taylor tail plus the selector defects $\mathsf{p}_{2n}-1$; on each of the other three blocks, where the nonzero target (if any) is matched by the other summand, the evaluated Taylor polynomial is finite with at most geometric coefficient growth, $|\mathcal{T}_{n,\bullet}| \le C(n+1)^5 M_\bullet (2/r_0)^n$ there, while at least one of its selector factors sits at its order-$2n$ zero and is bounded by $(4r(1+r))^{2n} \le (C/s)^{2n}$, which absorbs the growth after enlarging c_0; no convergence of the Taylor series outside its polydisc is used. Then Cauchy estimates bound the derivatives of order at most two on the real blocks by $CM_\bullet(c_{\mathrm{pol}}/s)^n$ with $c_{\mathrm{pol}} = 2^6 c_0$. By (E3), $\|\mathsf{D}_{\mu,s} - \mathsf{F}_n\| \le Cs^{-2}(1+\xi)^2 e^{-\xi/2}(c_{\mathrm{pol}}/s)^n$, which gives the convergence with $S_{\mathrm{pol}} = \max\{5, 2c_{\mathrm{pol}}\}$ (the bound (E3) requires $s \ge 5$; and since $c_0 \ge 3/r_0$, $s \ge 2c_{\mathrm{pol}}$ keeps the complex neighbourhoods of the blocks, of radius $3/s$ in the scaled variables, inside the polydisc of holomorphy), and multiplied by the rank of G_n gives (E5). For (E6), expand $q_{n+1,\bullet} - q_{n,\bullet}$ in the midpoint monomials $((\mathbf{x}-s\mathbf{e}_1/2)/s)^\alpha((\mathbf{y}-s\mathbf{e}_1/2)/s)^\beta$; the coefficient sums are at most $CM_\bullet K_{\mathrm{mid}}^n$ for a universal K_{mid}, because the selectors and shifts multiply coefficient sums by at most geometric factors. Lemma 9 at $b = \frac{1}{2}$, with $U = U_s \subset B_T(s\mathbf{e}_1/2)$, $T = s/2+1 \le s$, bounds each normalized moment $s^{-|\alpha|}\mathsf{m}_\alpha^\bullet$ by $M_{1/2}\,T(1+\mu T)^{-1}e^{\Lambda_0\mu T}B_{1/2}^{|\alpha|}$ for $\bullet = f$, and with $T^2(1+\mu T)^{-2}$ in place of $T(1+\mu T)^{-1}$ for $\bullet = g$, in the sense that the moment is $\langle\psi, \mathsf{R}_{U_s}^{1/2}\cdot\,\rangle$ for a vector ψ of at most that norm, so the increment is $\mathsf{R}_{U_s}^{1/2}\mathsf{B}_n\mathsf{R}_{U_s}^{1/2}$ with B_n a finite combination of rank-one operators of total trace norm at most $C(1+K_{\mathrm{mid}})B_{1/2}^9 M_{1/2}^2 e^{2\Lambda_0\mu T}(K_{\mathrm{mid}}B_{1/2}^9)^n$ times $[M_f T^2(1+\mu T)^{-2} + M_g T^4(1+\mu T)^{-4}] \le Ce^{-\xi/2}$, using $T \le s$ and $1+\mu T \ge (1+\xi)/2$. This is (E6) with $K_{\mathrm{pol}} = K_{\mathrm{mid}}B_{1/2}^9$. □

Proof of Theorem 5. Let $\theta = 1/(4\Lambda_0) = 2^{-17}$, so that $a = \theta/2$ and $\gamma = 2(1-\theta)$. For each increment, the analytic family $z \mapsto \mathsf{R}_{U_s}^{(1-z)/2}\mathsf{B}_n\mathsf{R}_{U_s}^{(1-z)/2}$ on $0 \le \operatorname{Re} z \le 1$ is continuous in trace norm and holomorphic inside (it is a finite combination of rank-one operators, and $w \mapsto \mathsf{R}_{U_s}^{w}$ is strongly continuous on $\operatorname{Re} w \ge 0$), with boundary trace norms $\|\mathsf{G}_n\|_1$ at $\operatorname{Re} z = 0$, where the imaginary powers are unitary, and $\|\mathsf{B}_n\|_1$ at $\operatorname{Re} z = 1$. Hadamard's three-lines theorem for the trace norm [33] at $z = \theta$ gives $Y_n = \mathsf{R}_{U_s}^{(1-\theta)/2}\mathsf{B}_n\mathsf{R}_{U_s}^{(1-\theta)/2}$ with $\|Y_n\|_1 \le \|\mathsf{G}_n\|_1^{1-\theta}\|\mathsf{B}_n\|_1^{\theta}$ and $\mathsf{G}_n = \mathsf{R}_{U_s}^{a} Y_n \mathsf{R}_{U_s}^{a}$. Inserting (E5) and (E6), with $2\Lambda_0\theta = \frac{1}{2}$, $s^{-2(1-\theta)} = s^{-\gamma}$ and $\mu T/2 \le 3\xi/8$ for $s \ge 4$,

$$\|Y_n\|_1 \le C_1^{1-\theta}C_2^{\theta}s^{-\gamma}(1+\xi)^2 e^{-\xi/8}(n+1)^3\big[(c_{\mathrm{pol}}/s)^{1-\theta}K_{\mathrm{pol}}^{\theta}\big]^n.$$

Choose $S_{\mathrm{rev}} \ge \max\{S_{\mathrm{pol}}, 5\}$ so large that the bracket is at most $\frac{1}{2}$ for $s \ge S_{\mathrm{rev}}$. Since $\sum_{n\ge0}(n+1)^3 2^{-n} = 52$ and $\sup_{\xi\ge0}(1+\xi)^2 e^{-\xi/8} \le 256$, the series $Y_{\mu,s} = \sum_n Y_n$ converges in trace norm with $\|Y_{\mu,s}\|_1 \le \mathcal{J}_0 s^{-\gamma}$, $\mathcal{J}_0 = 52\cdot 256\, C_1^{1-\theta}C_2^{\theta}$. Each Y_n is self-adjoint because B_n is, and $\mathsf{R}_{U_s}^{a}Y_{\mu,s}\mathsf{R}_{U_s}^{a} = \sum_n \mathsf{G}_n = \lim_N \mathsf{F}_{N+1} = \mathsf{D}_{\mu,s}$. □

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* D.T. built and ran the automated research and formalization workflows at PSI that produced these results, steered the writing of the paper, and asked for AI reviews and revisions of the drafts.

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